

ENVIRONMENT AND CLIMATE CHANGE: BUILDING A RESILIENT, COMPETITIVE AND DEVELOPMENT-DRIVEN INDIA



India aims to become a developed nation by 2047, which requires achieving high, inclusive, and environmentally sustainable growth. This will necessitate a transformation in consumption and production patterns, as well as technological and policy reforms. Mission LiFE (Lifestyle for Environment) emphasises the importance of behavioural changes and citizen participation in India's approach to climate action and sustainability, working alongside government-led mitigation and adaptation efforts.

Although India's per capita emissions remain well below the global average, climate change poses risks to livelihoods, infrastructure, and economic stability. Adaptation is, therefore, central to India's climate strategy, driven primarily by public investment and community-based action. The chapter examines how State Action Plans on Climate Change integrate water security, resilient livelihoods, urban resilience, and ecosystem restoration into development objectives, ensuring that development gains are sustained.

India is also pursuing a balanced mitigation pathway, scaling renewables, battery storage, and nuclear energy in line with the objective of energy security and industrial competitiveness. However, inadequate global capital flows to developing countries remain a significant constraint, underscoring the need to reform multilateral financial institutions and to strengthen domestic financial systems. The chapter further reviews the evolution of environmental regulations and their role in enabling sustainable development.

INTRODUCTION

10.1 The global climate change agenda has reached an inflexion point. What was once framed as a straightforward moral and technological transition toward a net-zero future is today marked by complex trade-offs, capacity constraints and a widening gap between ambition and operational reality. Nowhere is this tension more visible than in parts of Europe, which have presented themselves as the vanguard of the energy transition but are now confronting the practical and institutional stresses that follow from rapid decarbonisation undertaken ahead of grid readiness, storage depth, and industrial

system resilience.¹ These developments matter deeply for emerging economies such as India because they reveal the structural nature of transition risk, the importance of sequencing, and the consequences of allowing complexity to outpace institutional capacity.

10.2 Europe's recent experience illustrates these pressures in a particularly striking way. In the Netherlands, celebrated as a leader in renewables and electric mobility, the rapid deployment of solar and wind energy has collided with a distribution network designed for an earlier era of centralised thermal generation.² The result has been congestion, curtailment, and long waiting times for industrial connections, as well as appeals to households to moderate evening consumption to avoid overloading the system. The challenge here is not ideological, but infrastructural: renewable capacity has expanded faster than balancing systems, transmission reinforcement and storage technologies. The consequence is that intermittency-related costs — from maintaining thermal reserves to the capital burden of grid upgrades — are felt in both reliability stress and headline tariffs.

10.3 A similar story is visible elsewhere. On 28th April 2025, Spain³ experienced a widespread grid disruption triggered by a combination of network instability during a period of high renewable energy output and transmission system sensitivity under shifting load conditions. The result is that intermittency-related costs — from maintaining thermal reserves to the capital burden of grid reinforcement — increasingly surface not through higher tariffs but through episodes of reliability stress and tighter operating margins. This isn't criticism of renewable energy itself; it's a reminder that if we introduce complex systems too quickly without buffers, redundancy and institutional capacity, the system is likely to become fragile instead of thriving.

10.4 Taken together, these strands of evidence point to a common lesson: transitions are more durable when sequencing, system buffers and institutional depth are treated as intrinsic elements of design rather than assumed to follow. This is especially salient for emerging economies, where growth, energy security and resilience must advance alongside low-carbon pathways, and not be displaced by it.

10.5 Across several advanced economies, these stresses follow a common pattern: storage lags investment commitments, balancing costs rise, and transmission

1 Damen, M. (2023). Four challenges of the energy crisis for the EU's strategic autonomy (EPRS Briefing No. PE 747.099). European Parliamentary Research Service. Retrieved December 28, 2025, from <https://tinyurl.com/2kcea6p8>.

2 Laurenson, J. (2025, October 16). Netherlands' renewables drive putting pressure on its power grid. Bloomberg. Retrieved on December 20, 2025 from: <https://tinyurl.com/38bnb89j>

3 Bloomberg. (2025, July 16). Spain blackout investigation shows EU grids need more resilience. Bloomberg News. Retrieved December 29, 2025, from <https://tinyurl.com/4f342ahs>.

bottlenecks multiply. The result is what may be described as an entropy effect, in which well-intentioned transition strategies inadvertently increase systemic volatility unless accompanied by buffers, redundancies and stabilising structures.

10.6 This pattern underscores a broader lesson. In the early years after the Paris Agreement, there was a disproportionate emphasis on renewable energy capacity additions, procurement targets, and signalling ambition, while underestimating the engineering and financial challenges of maintaining stability in grids dominated by intermittent generation. The real cost of renewable power at scale includes storage, reserve capacity, balancing operations and transmission reinforcement. When these remain implicit rather than explicit, the policy narrative tends to understate the degree of institutional preparation required for a reliable and just transition.

10.7 More recently, the broader global climate discourse has undergone a gradual shift toward greater realism. Policymakers and assessment institutions have become more attentive to the distinction between what climate science can state with high confidence and what remains subject to uncertainty, natural variability and modelling limitations. The evidence from official assessments indicates that while the planet has warmed and human activity exerts a measurable warming influence, the translation of these findings into deterministic or catastrophic policy narratives often compresses nuance and downplays uncertainty. A more careful reading suggests that climate risk — though serious — is best addressed through measured, evidence-based strategies that preserve energy security, competitiveness and social stability, rather than through alarmist framings that may weaken institutional credibility or crowd out legitimate policy trade-offs.

10.8 A complementary body of empirical work⁴ points in a similar direction, underscoring the value of calibrated interpretation when translating climate evidence into policy and design. The Intergovernmental Panel on Climate Change (IPCC)'s 2021 release of local sea-level projections has been an important step for infrastructure and adaptation planning; however, a recent global comparison with tide-gauge observations suggests that only a small proportion of locations worldwide records of sufficient length and continuity for robust estimation. Across this subset, nearly 95 per cent of sites show no statistically significant acceleration in sea-level rise, and projected rates are, on average, about two millimetres per year higher than those observed, with the limited instances of acceleration largely associated with local, non-climatic factors, such as subsidence or land-use-driven ground movement.⁵ The implication is not to downplay

4 Voortman, H.G.; De Vos, R. A Global Perspective on Local Sea Level Changes. *J. Mar. Sci. Eng.* 2025,13, 1641. <https://doi.org/10.3390/jmse13091641> Retrieved on 20 January 2026 from <https://tinyurl.com/2xnb27pd>.

5 Voortman, H. G., & De Vos, R. (2025). A Global Perspective on Local Sea Level Changes. *Journal of Marine Science and Engineering*, 13(9), 1641. <https://doi.org/10.3390/jmse13091641> Retrieved on 20 January 2026 from <https://tinyurl.com/2xnb27pd>.

climate risk, but to highlight the need for context-specific assessment and institutional prudence in operationalising projections.

10.9 This more balanced perspective aligns with the argument advanced in Bill Gates' reflections ahead of the Conference of the Parties (COP)30.⁶ The central theme is that climate policy should prioritise human welfare, particularly for poorer and climate-vulnerable societies. Climate change will have meaningful consequences, but it is unlikely to produce civilisational collapse; innovation, declining green-cost premia and targeted investments are more durable low-carbon pathways than rapidly imposed constraints that erode growth. Equally, scarce fiscal resources should not be diverted away from health, agriculture and poverty reduction merely to accelerate near-term mitigation milestones. Growth and prosperity strengthen resilience and reduce vulnerability, and must therefore be treated as complementary to climate strategy rather than competing with it.

10.10 Taken together, these perspectives reinforce an important ethical and strategic principle: **development is, in itself, a form of adaptation**. While emissions mitigation is also important, effective climate action encompasses much more; it also depends on improving societal capacity to withstand shocks through stronger health systems, climate-resilient infrastructure, enhanced agricultural productivity, and access to affordable and reliable energy. Measured in terms of human welfare rather than temperature outcomes alone, the most effective strategies are those that expand opportunity while reducing vulnerability, remaining transparent about uncertainties and pragmatic about the limits of what climate science can presently resolve with confidence.

10.11 For India, these developments carry important implications. Achieving sustained growth and rising living standards will require a substantial expansion in the supply of affordable and reliable electricity. Renewable energy will play a major and growing role in this expansion; however, capacity additions alone do not automatically translate into a dependable supply, particularly at non-solar peak times or during periods of high variability. As renewable penetration rises, the cost of maintaining dispatchable thermal capacity increases, even as utilisation declines, while grid stability challenges intensify. India must, therefore, approach the coming decade not as a climate policy problem in isolation, but as a broader energy system strategy — one that sequences the transition in line with growth, security, and institutional preparedness.

10.12 Such a strategy must prioritise pragmatism over signalling: maintaining sufficient dispatchable power capacity to support industrialisation and social development even

6 Gates, B. (2025, October 28). Three tough truths about climate. Gates Notes. Retrieved December 29, 2025, from <https://tinyurl.com/mryd8455>.

as renewables scale; strengthening transmission and distribution; investing in storage and grid-management technologies; and restoring emphasis on hydro and nuclear as long-horizon anchors for low-carbon development. Internationally, India's negotiating stance must remain grounded in first principles — predictable and front-loaded finance reflecting full-system costs, technology flows that support resilience rather than compliance, and commitments that respect differentiated capabilities, responsibilities and risk exposures.

10.13 Advanced economies have been the first to discover that decarbonisation at scale is an engineering, financial and institutional undertaking with unavoidable trade-offs. Their experience does not diminish the urgency of climate action; rather, it clarifies the conditions under which it can be sustained. For India, the lesson is to articulate, calmly and firmly, a pathway that protects growth, builds resilience and preserves strategic flexibility, while remaining committed to responsible climate action on terms that are technologically feasible, developmentally coherent and compatible with India's economic aspirations.

10.14 India has grown sustainably with per capita carbon emissions considerably lower than the global average.⁷ India's climate action strategy has developed around three main pillars. Firstly, adaptation needs to be at the forefront given the country's significant vulnerability to physical impacts of climate change, such as the loss of life and property and health concerns due to intensification of extreme weather events, concerns of water and food security, slow-onset events like rising sea-level, amongst others.⁸ Prioritising adaptation not only addresses immediate climate challenges but also provides local economic benefits by protecting livelihoods, safeguarding infrastructure investments, and minimising disaster-related losses. This approach is largely realised through public investments initiated by the central government, strategic planning at the state level, and community-driven programs woven into broader development efforts.

10.15 Secondly, India is also taking mitigation action by phasing-in renewables, improving energy efficiency and diversifying energy sources to green hydrogen and nuclear power. The approach is to prioritise energy security, affordability and industrial competitiveness. However, this transition faces various intertwined challenges, including limited access to advanced technologies, vulnerabilities in supply chains for critical minerals essential for clean energy, and constrained availability of reliable and affordable financial capital.

7 India's per capita GHG emissions for 2024 have been estimated to be 2.9 t/person vis-à-vis GHG average of 6.7 t/person in Per capita greenhouse gas emissions. (2025, November 13). Our World in Data. Retrieved December 26, 2025, from <https://tinyurl.com/bdeazps2>.

8 Initial Adaptation Communication: National Communication (NC). NC 3. (2023). In UNFCCC. Retrieved December 26, 2025, from <https://unfccc.int/documents/636235>.

10.16 Thirdly, a predominant reliance on domestic resources as international climate finance and international financial regulation has both become binding constraints. While global capital is abundant, it does not flow at scale to developing countries due to structural features of the international financial system and risk perceptions, resulting in a high cost of capital. In recent years, India has made significant progress in enhancing its financial ecosystem through regulatory reforms. The country has streamlined its financial markets and introduced innovative instruments for sustainable finance, including frameworks for thematic bonds like green, blue, yellow, transition, social, sustainability to support sustainable development goals including interventions to address climate action projects. Sovereign green bonds have backed public sector initiatives focused on environmentally friendly projects, while a green deposit framework emphasises mobilising capital for sustainability. Additionally, specialised financial institutions are playing a vital role in aligning the financial sector with the nation's climate sustainability goals.

10.17 Internationally, there's a growing need to reform multilateral development finance,⁹ credit ratings,¹⁰ and the use of risk-sharing instruments.¹¹ At the same time, environmental regulation in India is transitioning from procedural, command-and-control approaches toward risk-based, technology-enabled, and outcome-oriented governance that supports both environmental protection and ease of doing business.

10.18 Against this backdrop, this chapter examines India's national and subnational strategies across adaptation, mitigation, climate finance, and regulation; it underscores the importance of Mission-LiFE as a cross-cutting theme and draws lessons for policy design and economic planning. It highlights how climate action, when embedded in development priorities, can strengthen resilience, enhance competitiveness, and support India's long-term growth trajectory.

10.19 The rest of the chapter is organised as follows: The second section focuses on adaptation measures, highlighting the actions taken by the Government of India and various subnational initiatives that address a range of local needs. The third section discusses mitigation strategies, including India's renewable energy imperatives, access to critical minerals, the political economy of the transition, and the new Carbon Credit Trading Scheme. The fourth section addresses Mission LiFE and its implementation

9 Lahn, G., & Schröder, P. (2023, September 21). If the SDGs are to survive, multilateral development banks must embrace reform. Chatham House. Retrieved December 26, 2025, from <https://www.chathamhouse.org/about-us/our-people/patrick-schroder>

10 Rating the globe: reforming credit rating agencies for an equitable financial architecture. (2025, March 6). United Nations University. Retrieved December 26, 2025, from <https://tinyurl.com/bdd763ha>.

11 Laxton, V., Alayza, N., & Neunuebel, C. (2025, June 29). 4 ways financing for development can deliver for people, nature and climate. World Resources Institute. Retrieved December 28, 2025, from <https://tinyurl.com/mr2am9xc>.

across different sectors. The fifth section covers climate finance, detailing the steps India is taking to mobilise resources for climate action. Finally, the sixth section examines the evolution of environmental regulations and the measures being implemented to enhance its effectiveness. The chapter concludes with key lessons and recommendations for future action.

Adaptation: Strengthening climate resilience

10.20 Given our country's vulnerability to climate change, action is necessary to ensure continuing economic stability, secure and enhanced livelihoods, and to protect infrastructure from risks such as droughts, floods, and rising sea levels. Integrating climate adaptation and resilience into development plans is essential for sustainable growth.¹² Unlike mitigation, adaptation offers immediate benefits by reducing losses, stabilising incomes, locking in development and safeguarding investments, helping us build a more resilient future.

10.21 Despite the urgency for adaptation, international and private capital flows remain constrained and disproportionately skewed towards mitigation efforts. The 2025 Adaptation Gap Report estimates that developing countries will require between USD 310 billion and USD 365 billion annually by 2035, whereas the current flow of international public adaptation finance to developing countries is approximately USD 26 billion.¹³ The Economic Survey for 2024–25 emphasised the importance of integrating climate adaptation into our policy-making process. This Chapter provides further details on this subject with specific reference to the sub-national measures, acknowledging that the challenges posed by climate change and the ways to tackle them are often unique to specific locations and communities.

Public Investment–Led Climate Adaptation in India

10.22 India's climate adaptation strategy is predominantly advanced through a development-led approach, utilising domestic public investment in core development sectors. Adequate and sustained financing is the backbone of effective adaptation and resilience-building in India. Scaling up existing adaptation actions and addressing the substantial financing gap requires diversified resource mobilisation. Reflecting a strategic pivot towards self-funded resilience, India mainstreams adaptation through national and state-level planning backed by domestic resources. India's adaptation and resilience-related domestic spending surged from 3.7 per cent of the GDP in FY16

12 IMF. (2025). India 2025 Article IV consultation: Press Release; Staff Report; Staff Statement; And Statement by the Executive Director for India. Retrieved December 26, 2025, from <https://tinyurl.com/ye6mm956>.

13 'United Nations Environment Programme. (2025). Adaptation gap report 2025: Running on empty - The world is gearing up for climate resilience — without the money to get there. Retrieved on December 15, 2025, from <https://tinyurl.com/mrcpsm2a>.

to 5.6 per cent of the GDP in FY22.¹⁴ The National Action Plan on Climate Change (NAPCC)¹⁵ spearheads climate action through nine missions – on solar energy, energy efficiency, sustainable habitats, water management, the Himalayan ecosystem, green India, sustainable agriculture, and strategic knowledge for climate change and health – implemented by the respective nodal ministry/department. Many of these are focused on adaptation.

10.23 The National Mission on Sustainable Agriculture¹⁶ promotes climate-resilient farming through initiatives such as "Per Drop More Crop," which focuses on efficient water use, Rainfed Area Development for integrated farming, and Soil Health Management supported by the Soil Health Card scheme. In the area of water management, the National Water Mission emphasises conservation and fair access through integrated resource management. It supports State-Specific Plans and rainwater harvesting initiatives, providing financial help to States and Union Territories. The National Programme on Climate Change and Human Health raises awareness about health impacts from climate change and aims to strengthen health systems, particularly for vulnerable groups. The National Mission on Sustainable Habitat integrates adaptation into urban development through programs such as the Swachh Bharat Mission and urban transport initiatives, thereby enhancing climate resilience and liveability.

10.24 India has adopted an integrated approach to coastal and marine resilience, linking ecosystem protection with livelihood security and climate adaptation. The National Coastal Mission strengthens integrated coastal zone management and climate-resilient infrastructure, thereby directly reducing the vulnerability of coastal communities to sea-level rise and extreme weather events. Complementing this, the Mangrove Initiative for Shoreline Habitats & Tangible Incomes (MISHTI) envisages the restoration and reforestation of mangroves over approximately 540 square kilometres across nine coastal States and four Union Territories over a five-year period (2023–2028). The programme is implemented through convergence with existing schemes and programmes, enabling states to integrate mangrove restoration within ongoing development efforts. MISHTI is expected to generate around 22.8 million person-days of employment, create an estimated carbon sink of 4.5 million tonnes, and unlock opportunities for nature-based tourism and sustainable livelihoods for coastal communities.¹⁷ In parallel, the National Plan for Conservation of Aquatic Ecosystems

14 Initial Adaptation Communication: National Communication (NC). NC 3. (2023). In UNFCCC. Retrieved December 26, 2025, from <https://unfccc.int/documents/636235>.

15 Press Information Bureau. (2025, July 24). Parliament Question: - Measures related to climate change. Retrieved on December 15, 2025 from <https://tinyurl.com/2tph4nj3>.

16 Department of Agriculture & Farmers Welfare. (2025). Rainfed farming system: National Mission for Sustainable Agriculture (NMSA). Government of India. Retrieved December 28, 2025, from <https://tinyurl.com/3ymvyym>.

17 Press Information Bureau. (2023, June 05). World Environment Day 2023 celebrated with a thrust on Mission LiFE. Retrieved on December 24, 2025, from: <https://tinyurl.com/4c2vwdje>.

has expanded protected wetlands that support flood moderation and water security—critical buffers under climate stress. Together with Mission LiFE, which promotes behavioural adaptation and mindful resource use, these initiatives reflect a shift toward ecosystem-led, development-integrated climate resilience.

10.25 In India's federal framework, climate adaptation is advanced through coordinated action across national, state, and local levels. National frameworks and programmes provide policy coherence, financial support, and institutional mechanisms, while States contextualise and operationalise these interventions through sectoral policies, public programmes, and local institutions. These have played a critical role in enhancing development and environmental conservation activities that promote ecological restoration and strengthen rural-urban infrastructure. The Viksit Bharat – Guarantee for Rozgar and Ajeevika Mission (Gramin) (VB-G RAM G) Act, 2025, aims to strengthen the interventions. Details are covered in **Chapter XIII Rural Development**.

Box X.1: Kerala Climate Resilient Agri-Value Chain Modernisation Project (KERA)¹⁸

The Government of Kerala, with support from the World Bank, is implementing the Kerala Climate-Resilient Agri-Value Chain Modernisation (KERA) Project to build a climate-resilient, competitive, and market-oriented agricultural sector. With an outlay of USD 285 million (₹2,365 crore), the project aims to enhance climate resilience, boost productivity, promote value addition, and strengthen market linkages across priority agri-value chains, directly benefiting over 400,000 farmers statewide.

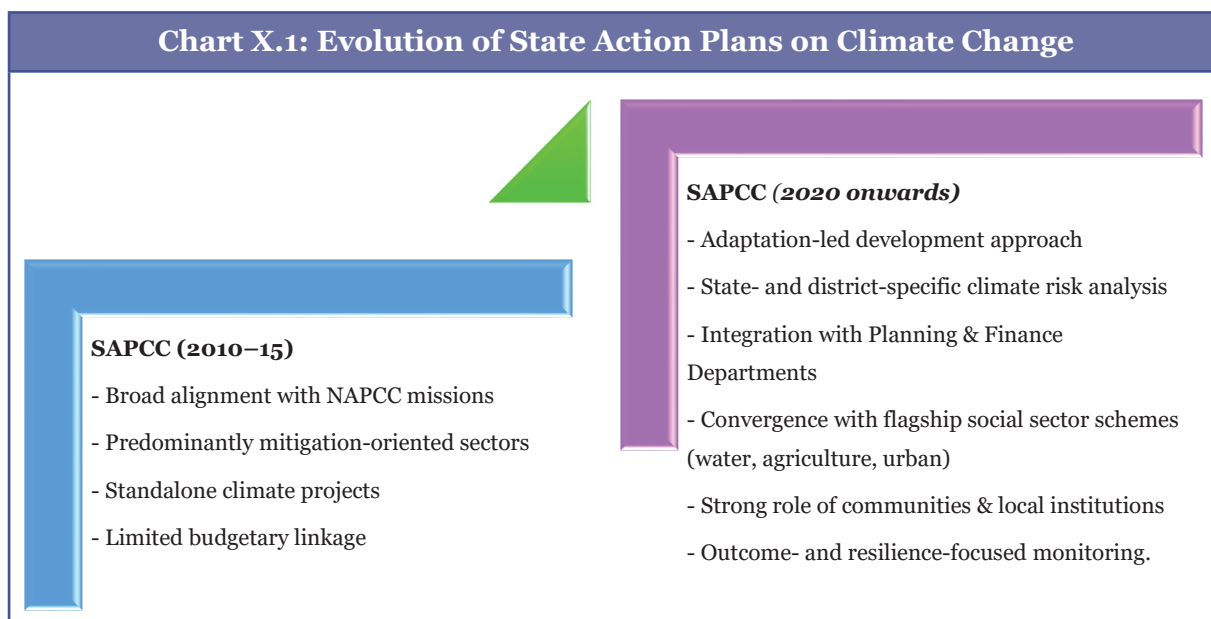
KERA is structured around three core components. First, it promotes climate-resilient and low-carbon cultivation processes as a co-benefit of mitigation. Second, it supports smallholder commercialisation by facilitating up to 150 Productive Alliances between Farmer-Producer Companies (FPCs) and agribusinesses, with grants of up to ₹2 crore per alliance, alongside support for replantation of certified climate-resilient varieties of coffee, cardamom, and rubber. Third, the project strengthens agri-food MSMEs and agri-tech start-ups through grants to about 200 MSMEs (up to ₹1.5 crore each), support to 150 agri-tech start-ups, and the development of agri-parks and food parks to enable cluster-based growth.

Implemented across all 14 districts of Kerala, KERA focuses on measurable outcomes—higher farm incomes, enterprise growth, job creation, and improved resilience to climate shocks—while ensuring inclusive participation of small and marginal farmers, women-led enterprises, FPCs, and rural entrepreneurs.

10.26 Further, states play a crucial role in planning and implementing adaptation strategies to address climate change. The State Action Plans on Climate Change (SAPCCs) are crucial tools for translating NAPCC's broader objectives into actionable steps. These plans are tailored to the specific needs of different regions, whether they are

¹⁸ Inputs received from the Government of Kerala.

vulnerable coastal areas or unique Himalayan ecosystems. Initially, the SAPCCs placed a strong emphasis on mitigation efforts to reduce emissions. Recent revisions of SAPCCs¹⁹ reflect a gradual shift towards adaptation-led development, with increasing emphasis on contextual risk considerations, sectoral vulnerabilities, and alignment with district-level development priorities, where available. Overall, it is about making our responses to climate change more robust and better aligned with the unique challenges faced by each region.



Subnational initiatives on adaptation through an ecosystem-based approach and Institutional Innovations

10.27 The case studies depicted below illustrate how adaptation has evolved from stand-alone projects to integrated, growth-supporting national resilience strategies.

10.28 Meghalaya – Ecosystem-Based Water Security and Climate Adaptation²⁰: Meghalaya is experiencing increasing concerns about water security, with water flow at nearly half of the state's springs either dried up or showing a significant reduction in water discharge. To address this, the state has embraced a community centric adaptation strategy, which includes interventions such as the Protection of Vulnerable Catchment Areas in Meghalaya (MegARISE), the Spring Mapping Initiative, and the Climate-Adaptive Community-Based Water Harvesting Project. The MegARISE project aims to enhance water availability and security by protecting critical catchment areas

19 Kerala - <https://envt.kerala.gov.in/wp-content/uploads/2022/12/Kerala-State-Action-Plan-on-Climate-Change-2.0.pdf>. Punjab - https://ncdc.mohfw.gov.in/wp-content/uploads/2025/01/24_SAPCCHH_Punjab_21-10-24.pdf. Uttar Pradesh - https://upccce.org/public/UPLOADS/REPOSITORY/DOC/27_stateactionplan.pdf.

20 Inputs provided by the Government of Meghalaya.

through sustainable forest management. The initiative focuses on watershed treatment, forest restoration, and participatory community involvement, with plans to establish plantations over 8,430 hectares and to treat and protect two key catchments—Umiew and Ganol. Advanced Geographic Information System (GIS) mapping has been utilised to capture more than 55,000 springs, to enable timely corrective actions. Over 500 community-based water-harvesting projects are also being established to improve climate-resilient water management for communities in the state. These initiatives not only aim to secure water resources but also empower communities to adapt to the changing climate effectively. Collectively, these measures strengthen water security while enhancing community resilience to climate change in the state.

10.29 Odisha – Institutionalised Climate Adaptation through Water Governance²¹: Central to this effort are over 39,000 Pani Panchayats, which empower communities to manage water resources while ensuring the inclusion of women and marginalised users. The Odisha Integrated Irrigation Project for Climate Resilient Agriculture enhanced 538 minor irrigation systems across 15 districts, benefiting around 125,000 households. By integrating modern irrigation techniques with sustainable agriculture and aquaculture practices, the project has achieved a 28.8 per cent increase in water productivity, a 67.8 per cent rise in agricultural output, a 105 per cent growth in high-value vegetable farming, and a 27 per cent boost in household incomes. These initiatives are transforming lives and building a sustainable future for communities in Odisha.

10.30 Tamil Nadu’s Coastal Restoration Mission²²: Tamil Nadu has a 1,069 km coastline, home to 14 coastal districts that are vulnerable to impacts such as shoreline erosion, storm surges, and sea level rise. The Tamil Nadu Sustainably Harnessing Ocean Resources and Blue Economy Project employs a multi-pronged approach to enhance coastal biodiversity, improve coastal protection, enhance livelihoods, mitigate pollution, and enhance project management.

10.31 Ahmedabad - Heat insurance to safeguard informal workers’ wages: Ahmedabad became the first city in South Asia to launch a Heat Action Plan in 2013. Local grassroots organisations, such as the Mahila Housing Trust²³ and the Self-Employed Women’s Association,²⁴ have launched parametric heat insurance schemes to compensate women in the informal sector for wage losses on extremely hot days. The data-driven model uses real-time temperature monitoring to trigger automatic

21 Inputs provided by the Government of Odisha.

22 Inputs provided by the Government of Tamil Nadu.

23 The All India Disaster Mitigation Institute. (2025, May 14). Climate risk insurance: What worked! - The All India Disaster Mitigation Institute. Retrieved on December 23, 2025 from: <https://aidmi.org/blog/climate-risk-insurance-what-worked/in-inputs-provided-by-council-on-energy-environment-and-water-ceedw/>.

24 Nanavaty, R., & Saxena, P. (2025, March 25). How heatwaves have sparked new ways for women farmers in India to protect their crops. World Economic Forum. Retrieved December 24, 2025, from <https://tinyurl.com/mwf3ehtt> in inputs provided by Council On Energy, Environment and Water (CEEW).

payouts when the maximum temperature crosses certain predefined thresholds. The scheme charges an affordable annual premium of ₹354, and subscribers receive payouts ranging from ₹750 to ₹1,250.²⁵ By directly linking compensation to objectively measured weather conditions, the scheme provides a timely and predictable safety net for workers and recognises heat stress as an economic shock, rather than merely a public health concern.

10.32 Uttarakhand - A community radio station is leading disaster preparedness: Mandakini ki Aawaz, a community-run radio station in Sena Gadsari village, Rudraprayag, Uttarakhand, makes critical information and early warnings accessible to the community. By regularly corresponding with the India Meteorological Department, Dehradun, and the Uttarakhand State Disaster Management Authority, the radio station delivers early warnings, public service announcements, and disaster advisories that enhance preparedness before an extreme event, facilitate coordination during emergencies, and support recovery efforts.²⁶

10.33 Jodhpur - Community cooling stations are a model of community resilience: Jodhpur Nagar Nigam North and Mahila Housing Trust, supported by the Natural Resources Defence Council, recently launched a net-zero cooling station as part of the city's Heat Action Plan²⁷ in key heat-vulnerable localities in the city. The stations combine passive cooling (wind towers, vetiver curtains, misting fans) with solar power and reflective roofing. They provide drinking water and first aid, offering refuge for outdoor workers and daily wage earners, who have limited access to air-conditioned spaces. By integrating passive cooling techniques, renewable energy, and traditional architectural elements, the facility serves as a model for scalable urban community cooling infrastructure. Early evidence suggests that the conditions inside the cooling centre are substantially cooler than those outdoors. In late April 2024, the temperature inside was about 8°C cooler during the hottest hours, from 12 PM to 6 PM.²⁸

10.34 While these measures reflect a few examples of several such initiatives, climate resilience and adaptation can no longer be limited to managing forests, water bodies,

25 Dhruval Parekh, & Surangya Kaur. (2025, June 16). Beyond the thermometer: Ahmedabad uses cool roofs, misting bus stops & heat insurance for climate resilience. Down to Earth. Retrieved on December 23, 2025 from: <https://tinyurl.com/ypkfr2zh> in inputs provided by Council On Energy, Environment and Water (CEEW).

26 Khup Hangzo, PK (2023, October 19). Mandakini Ki Awaaz: Saving Lives, One Radio Broadcast at a Time. Vivekananda International Foundation, and Shawn Sebastian (2022, October 22). Mandakini Ki Awaz -How a community radio station is leading disaster preparedness in Uttarakhand in inputs provided by Council on Energy, Environment and Water (CEEW). Retrieved on December 22, 2025 from: <https://www.vifindia.org/print/12087> and <https://tinyurl.com/yruxjvbu> respectively.

27 Jodhpur Heat Action Plan in inputs provided by Council on Energy, Environment and Water (CEEW). Retrieved on December 22, 2025 from: <https://tinyurl.com/msbsa8b7>.

28 Vijay Limaye (2024, June 7). Jodhpur, India Unveils a Net-Zero Public Cooling Station. Natural Resources Defense Council in inputs provided by Council on Energy, Environment and Water (CEEW). Retrieved on 22 December 2025 from: <https://tinyurl.com/mtzdye82>.

or coastlines alone; they must be built into the economic and business frameworks that sustain livelihoods and cities. Services such as decentralised renewable energy, solid waste management, and safe drinking water are emerging as resilience assets, strengthening local adaptive capacity while also advancing India's mitigation goals. The Climate Resilient Villages initiatives, such as those in Karnataka,²⁹ Tamil Nadu,³⁰ and Odisha³¹ demonstrate how resilience can be institutionalised at the grassroots by transforming resilience from a project into a local governance function. As Indian cities continue to grow at a rapid pace, internalising climate risk into the fabric of urban planning, means considering how climate change affects land use, infrastructure, and the services provided to residents. By acknowledging these challenges, such as rising temperatures, flooding, and water shortages, we can better prepare our urban areas to address them. Building resilience is also vital for ensuring that the urban centres can thrive and continue driving economic growth.

Mitigation: Transition to a low-carbon economy

10.35 India is adopting a multifaceted approach to mitigate global warming by diversifying its energy sources and enhancing access while also increasing the share of non-fossil fuels, improving energy efficiency, and promoting stability across its energy systems. These strategies align with the nation's development and sustainability objectives. A comprehensive range of policies has been implemented to achieve the 2030 Nationally Determined Contributions (NDC) targets.

10.36 The Economic Survey 2024–25 noted the potential of a rapid expansion of variable renewable energy sources without adequate firm baseload capacity and grid reinforcement leading to higher electricity prices, thereby compromising energy security. Experiences in the United Kingdom highlight how declining firm power source has contributed to higher system costs. Another case is Germany, which is experiencing rising household and industrial energy prices as a result of its energy transition policies.³² As mentioned earlier, Spain and the Netherlands have recently faced growing grid instability and capacity limitations, reflecting the risks associated with transitions that outpace investments in baseload generation, transmission, and system flexibility. In the Netherlands, grid congestion has become severe enough to prompt public appeals to reduce electricity use during peak hours, while nearly 14,000 firms remain on waiting

29 Press Information Bureau. (2025, April 02). *Climate Resilient Coastal Fishermen Villages Programme*. Retrieved on December 24, 25 from: <https://tinyurl.com/mrxh8une>.

30 Kannan, V. and Chakraborty, A. (2025, April 02). *Sustaining Tomorrow: Locally Led Adaptation for a Changing Climate*. World Resource Institute India. Retrieved on December 20, 2025 from: <https://tinyurl.com/57dhkcxz>.

31 Inputs received from the Government of Odisha.

32 Bothe, D., Reichenbach, J., Bieniasch, M., Braendle, G., Coordt, L., Janssen, M., Kuttler, F., Lorenczik, S., Stoll, J., Begemann, U., Frewer, T., Branconi, C., Frontier Economics, & Melnikov, H. (2025). *NEW PATHWAYS FOR THE ENERGY TRANSITION ('PLAN B')*. In German Chamber of Commerce and Industry, Frontier Economics. Retrieved on January 12, 2026 from <https://tinyurl.com/c2hn3z3a>.

lists for new connections to the grid.^{33, 34} These cases underscore the importance of grounding energy transitions in baseload adequacy and system reliability to ensure secure and affordable power.

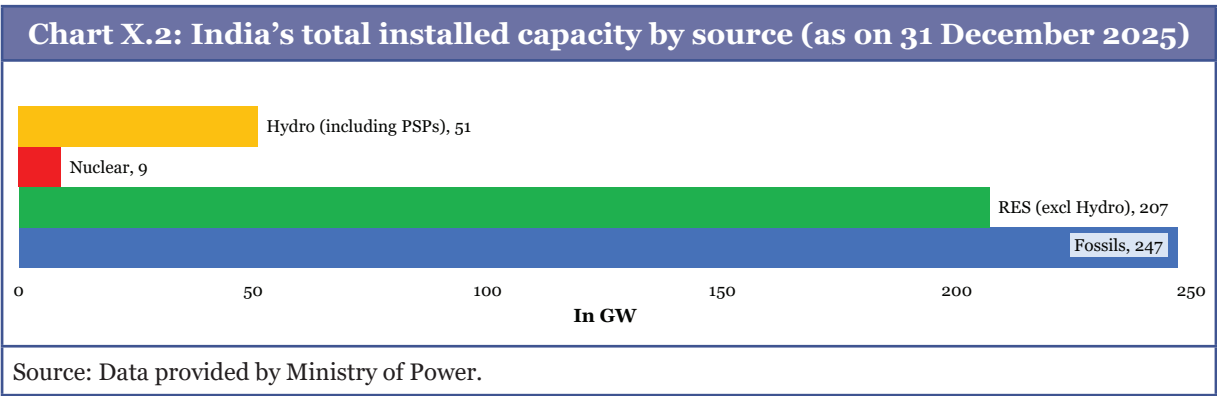
Energy Transformation at Scale: India's Renewable Imperative

10.37 India’s energy transition is being pursued through a combination of initiatives across various sectors, including nuclear, solar, and wind energy, green hydrogen, battery storage, and critical minerals, which help address both energy security and transition imperatives simultaneously. Details of some of these initiatives are presented in this section.

Recent measures to boost non-fossil fuel sources of energy

10.38 The country has already surpassed the goal of 50 per cent installed power capacity from non-fossil fuel sources (stood at 51.93 per cent at the end of December 2025³⁵), supported by record annual additions of renewable energy capacity (Chart X.2). During 2025-26 (up to 31st December 2025), a total of 38.61 GW of renewable energy capacity has been installed in the country, which includes 30.16 GW of solar power, 4.47 GW of wind power, 0.03 GW of Bio-Power and 3.24 GW of hydro power.³⁶

10.39 According to the International Renewable Energy Agency (IRENA)’s Renewable Energy Statistics 2025,³⁷ India now ranks fourth globally in total installed renewable energy capacity (after China, the USA, and Brazil), which underscores India’s growing influence in global clean energy markets.



33 Pascoe, R. (2025, October 6). 14,000 companies wait for a new Dutch power grid connection. DutchNews.nl. Retrieved on December 20, 2025, from: <https://tinyurl.com/2m7h4n6s>.

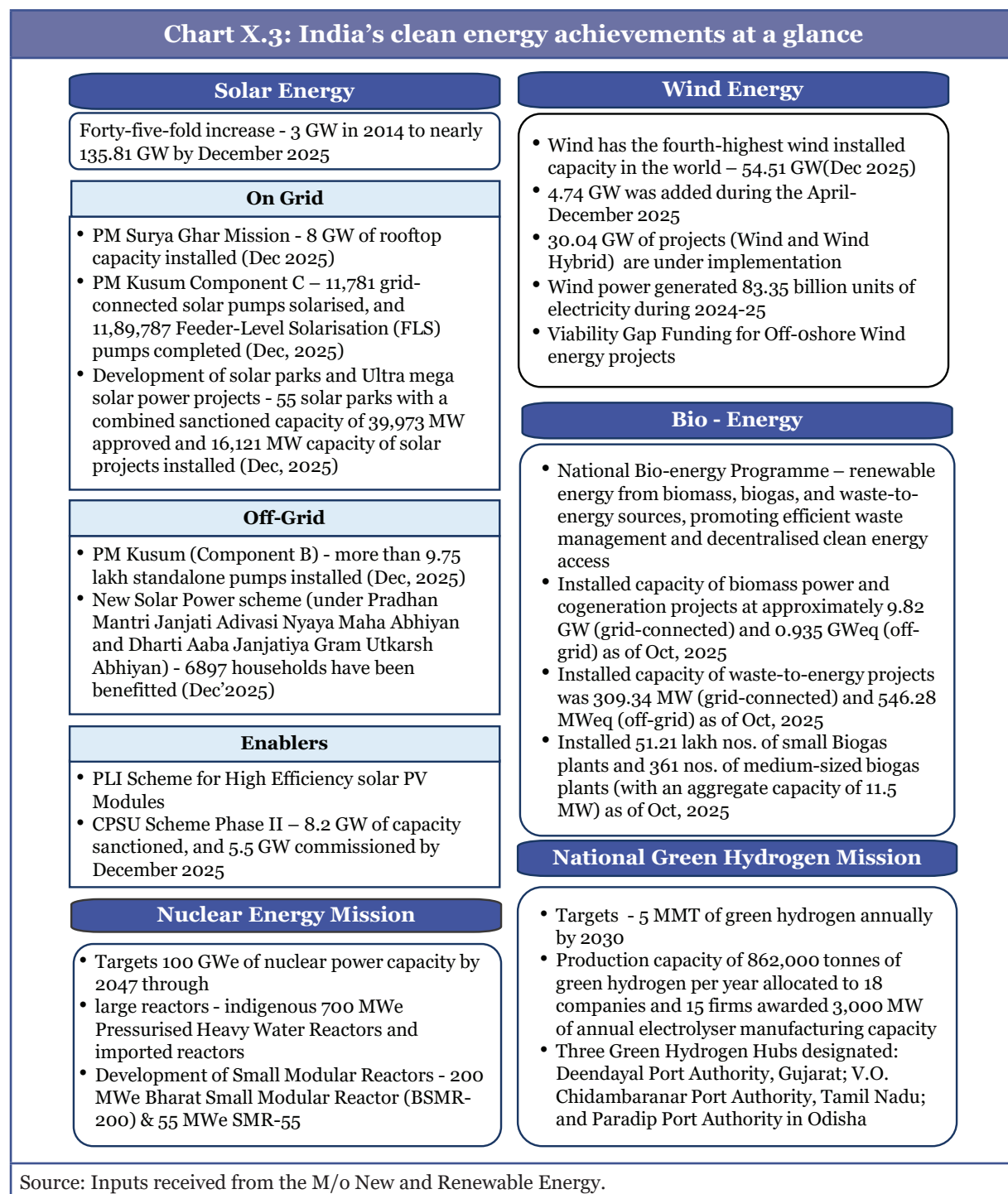
34 Laurenson, J. (2025, October 16). Netherlands’ renewables drive putting pressure on its power grid. Bloomberg. Retrieved on December 20, 2025 from: <https://www.bbc.com/news/articles/cn40y9yxkgvo>.

35 Data provided by Ministry of Power.

36 Inputs provided by Ministry of New and Renewable Energy.

37 IRENA Renewable Energy Statistics 2025. (2025, March 1). Retrieved on December 22, 2025 from: <https://tinyurl.com/55mybyte>.

10.40 The progress in expanding non-fossil fuel-based power capacity has been supported by a wide variety of initiatives to boost renewable energy systems, including solar, wind, and bio-energy. Additionally, new measures are being taken to support other clean sources of energy, including the National Nuclear Mission, the Green Hydrogen Mission, and the Bio Energy Programme.³⁸ A snapshot of India's key clean energy initiatives and achievements is shown in Chart X.3.



³⁸ Inputs received from the M/o New and Renewable Energy.

Box X.2: The Sustainable Harnessing and Advancement of Nuclear Energy for Transforming India (SHANTI) Act, 2025

Nuclear is one of the cleanest forms of energy, capable of overcoming the concerns of intermittency and energy security associated with other renewable sources of energy like solar and wind power. Besides, nuclear energy can provide a reliable energy source for heavy industries that find it difficult to use renewable sources due to technological challenges. It can be used to produce hydrogen to power vehicles, and industrial process.³⁹ The Government of India had announced the Nuclear Energy Mission in the Union Budget 2025-26 and allocated ₹20,000 crore with the aim of developing at least five indigenously designed and operational small modular reactors (SMRs) by 2033.⁴⁰

India currently has a total nuclear capacity of 8,780 megawatt (MW).⁴¹ The Central Electricity Authority has also estimated that the capacity mix of the power system in 2047 would have 100 GW of nuclear power capacity.⁴²

Therefore, as a major step forward in this direction, India has adopted the landmark Sustainable Harnessing and Advancement of Nuclear Energy for Transforming India (SHANTI) Act in December 2025. The SHANTI Act consolidates and modifies India's nuclear legal framework⁴³- the Atomic Energy Act, 1962, and the Civil Liability for Nuclear Damage Act, (CLNDA) 2010 to enable participation of the private sector or state governments, previously not allowed under the Atomic Energy Act, 1962 and also addresses liability structure under the CLNDA⁴⁴ by establishing a graded liability framework, but without diluting victim compensation.⁴⁵ The new framework enables private sector participation in key activities, including plant operations, power generation, equipment manufacturing and carrying out research and innovations in the field of peaceful uses of atomic energy.

10.41 Despite the progress in expanding non-fossil fuel energy, challenges remain. The renewable energy systems of solar and wind are highly material-intensive and require capital-intensive energy storage technologies for integrating renewable energy

39 Central Electricity Authority. (2025, June). Road map for achieving the goal of 100 GW of nuclear capacity by 2047 (Final Report). Ministry of Power, Government of India. Retrieved December 28, 2025, from <https://tinyurl.com/yuzryce7>.

40 Government of India, Ministry of Finance, Central Budget Division. (2025). Budget 2025-26 speech. <https://tinyurl.com/528s7wjs>.

41 Ministry of Power. (2025, November). Power Sector at a Glance: All India. Government of India. Retrieved December 28, 2025, from <https://tinyurl.com/4hkz29cs>.

42 Central Electricity Authority. (2025, June). Road map for achieving the goal of 100 GW of nuclear capacity by 2047 (Final Report). Ministry of Power, Government of India. Retrieved December 28, 2025, from <https://tinyurl.com/yuzryce7>.

43 Press Information Bureau. (2025, December 19). The Sustainable Harnessing and Advancement of Nuclear Energy for Transforming India (SHANTI) Bill, 2025 (Explainer ID 156593). Government of India. Retrieved December 28, 2025, from <https://tinyurl.com/3h7r9fa8>.

44 Press Information Bureau. (2025, December 17). Dr. Jitendra Singh says SHANTI Bill retains strong safety and liability safeguards amid Lok Sabha debate [Press release]. Government of India. Retrieved December 28, 2025, from <https://tinyurl.com/2pcdf7vv>.

45 Press Information Bureau. (2025, December 18). Rajya Sabha passes SHANTI Bill 2025, after it was passed by Lok Sabha [Press release]. Government of India. Retrieved December 28, 2025, from <https://tinyurl.com/2tyvukx8>.

into the energy grid. Material and storage requirements represent the two roadblocks to greater utilisation of these energy sources. The former extends beyond the access to critical minerals and is likely to have implications for the requirements of mining and energy for material processing (Box X.4). With respect to energy technology, the challenge is not just the need for new technologies, but also the substantial investment requirements. The unit cost of storage capacity installed is determined by the number of times each MW of storage is used in a year, which is likely to be less for short-term storage capacities than long-term capacities.⁴⁶ The following subsections delve into these two challenges.

The Case for Battery Storage

10.42 The Central Electricity Authority (CEA) estimated that India will require around 336 gigawatt-hours (GWh) of energy storage capacity by 2029–30 and 411 GWh by 2031–32 to support the reliable integration of renewable energy sources.⁴⁷ To enable this scale-up, a coordinated set of policy, regulatory, demand-side, and supply-side measures is being implemented.

10.43 Energy storage systems are formally recognised under the Electricity Rules as an integral part of the power system⁴⁸ and are included in the Harmonised Master List of Infrastructure⁴⁹ improving access to long-tenure, lower-cost finance. Storage has been embedded in the Resource Adequacy Planning Guidelines⁵⁰ and supported through a National Framework for the Promotion of Energy Storage Systems.⁵¹

10.44 Market and regulatory measures have been introduced to expand revenue streams for storage, including the waiver of inter-state transmission charges for co-located battery storage energy system (BESS) and pumped storage projects until June 2028,⁵² eligibility of storage for ancillary services,⁵³ participation in the High-Price Day-Ahead

46 Upadhyaya, Y., & Agrawal, M. (2025, March 29). Long term electricity storage is wishful thinking. Medium. Retrieved December 26, 2025, from <https://tinyurl.com/4nn7m5z7>.

47 Ministry of New and Renewable Energy. Energy Storage Systems (ESS) Overview. Retrieved on December 24, 2025, from: <https://tinyurl.com/3enfzypn>.

48 The Energy Conservation (Amendment) Act, 2022. Retrieved on January 20, 2026 from <https://tinyurl.com/37t99jwd>.

49 Infrastructure In India Initiative by Infrastructure Finance Secretariat, Department of Economic Affairs, Ministry of Finance. Retrieved on December 24, 2025, from: https://infrastructureinindia.gov.in/harmonized_list.

50 Ministry of Power (2023, June 28). Guidelines for Resource Adequacy Planning Framework for India-reg. Retrieved on December 24, 2025, from: <https://tinyurl.com/339u8uya>.

51 Ministry of New and Renewable Energy. National Framework for Promoting Energy Storage Systems by Ministry of Power. Retrieved on December 24, 2025, from: <https://tinyurl.com/3rsdnvma>.

52 Ministry of New and Renewable Energy. Waiver of Inter-State Transmission Charges for Energy Storage Systems (ESS). Retrieved on December 24, 2025, from: <https://tinyurl.com/3jw29mrn>.

53 Central Electricity Regulatory Commission. Central Electricity Regulatory Commission (Ancillary Services) Regulations, 2022. Retrieved on January 20, 2026 from <https://tinyurl.com/3tunwwud>.

Market,⁵⁴ and tariff-based competitive bidding guidelines for BESS procurement.⁵⁵ To accelerate deployment, the Government has launched two Viability Gap Funding (VGF) schemes supporting approximately 43 GWh of BESS (March 2024 and June 2025).⁵⁶ Further, manufacturing is supported through a ₹18,100 crore Production-Linked Incentive (PLI) scheme for 50 GWh of Advanced Chemistry Cell capacity, in which 10 GWh is earmarked for grid-scale storage.⁵⁷ For pumped storage projects, a grant-based support is provided to enable infrastructure.⁵⁸ Regulatory clearances have been streamlined for closed-loop projects, and ownership of storage has been liberalised to include consumers. Additional measures include advisory norms for co-locating storage of at least 10 per cent of installed solar capacity⁵⁹ to improve the dispatchability of solar power.

10.45 Recently, the International Finance Corporation (IFC)⁶⁰ committed USD 51.4 million (₹460 crore) in FY26 to IndiGrid Infrastructure Trust, an AAA-rated transmission infrastructure investment trust regulated by SEBI, to support the deployment of a 180MW/360Mwh utility-scale standalone BESS in Gujarat. The project is the largest standalone BESS in India, setting a scale and technology standard for future deployments. To de-risk this first-of-a-kind project and effectively reduce upfront costs, the commitment was strategically structured to include concessional support from the Clean Technology Fund (CTF), a multi-donor trust fund.

Box X.3: Renewable energy integration in the State of Kerala

Given the intermittent nature of solar energy, an effective way to utilise it would be to consume the electricity generated during the day and store the excess or export it to the grid. Batteries are still expensive; therefore, exporting to the grid remains a practical option, where a distribution company (discom) or power utility serves as the backup. To enable effective integration with the grid the Kerala State Electricity Board (KSEB)⁶¹ now requires new connected households,

54 Press Information Bureau. (2025, December 18). Development and Deployment of Energy Storage Capacities to Power Reliable Renewable Future. Retrieved on December 24, 2025, from: <https://tinyurl.com/ypkzbz4z>.

55 Ministry of Power. Guidelines for Procurement and Utilization of Battery Energy Storage Systems as of Generation, Transmission and Distribution assets, along with Ancillary Services. Retrieved on January 20, 2026 from <https://tinyurl.com/55f8x3cj>.

56 Press Information Bureau. (2025, December 18). Development and Deployment of Energy Storage Capacities to Power Reliable Renewable Future. Retrieved on December 24, 2025, from: <https://tinyurl.com/ypkzbz4z>.

57 Press Information Bureau. (2025, December 18). Development and Deployment of Energy Storage Capacities to Power Reliable Renewable Future. Retrieved on December 24, 2025, from: <https://tinyurl.com/ypkzbz4z>.

58 Ministry of Power. (2024, September 30). Modification of the scheme of Budgetary Support for the cost of Enabling Infrastructure for Hydro Electric Projects. Retrieved on December 24, 2025, from: <https://tinyurl.com/jc9pme2m>.

59 Central Electricity Authority (2025, February 18). Advisory on co-locating Energy Storage Systems with Solar Power Projects to enhance grid stability and cost efficiency reg. Retrieved on December 24, 2025, from: <https://tinyurl.com/374xw9dv>.

60 Inputs provided by International Finance Corporation.

61 Kerala State Electricity Regulatory Commission. (2025, November 05). Kerala State Electricity Regulatory Commission (Renewable Energy and Related Matters) Regulations, 2025. Retrieved on December 24, 2025, from: <https://tinyurl.com/2tz3bkfk>.

as well as industrial and agricultural solar plants, to install battery storage capacities. For example, Rooftop solar systems above 10 kilowatt(kW) would need to have 10 per cent battery storage, and those between 15 and 20 kW would require 20 per cent. After 2027, even smaller 5 kW systems would require storage. Battery capacities in solar plants are expected to address the issue of intermittency of solar power. Another incentive to increase the uptake of battery storage alongside solar power is also being provided through a gross metering mechanism, which offers higher tariffs to customers who have battery storage.

Critical Minerals and the Political Economy of the Energy Transition- Balancing Resilience, Standards, and Inclusiveness

10.46 The global energy transition is no longer solely determined by technology; it is increasingly constrained by who controls critical minerals. Metals like Lithium, cobalt, nickel, copper, and rare earth elements have become the new strategic chokepoints in shaping the contours of a low-carbon economy, influencing energy security, industrial competitiveness, and geopolitical power, as observed through several trade restrictions on export of critical minerals by source countries.⁶² Copper, in particular, is becoming a highly price-volatile metal⁶³ due to a number of factors. These include a series of mine outages⁶⁴ in Indonesia, Congo, and Chile⁶⁵ rising concerns of a supply deficit in medium to long terms, given perpetually growing demand from the power sector and data centres across the world, and trade protectionist measures.⁶⁶

Box X.4: Material requirements for Renewable Energy

The numbers below give a sense of the amount of investment that India may be required to make in renewable energy generation and the tremendous amount of energy they need in the first place:

Solar panels with a power capacity of 1 GW require approximately 18.5 tons of silver, 2,000-3,000 tons of polysilicon, and 10,252 tons of aluminium. To make the 1 GW, 18.5 tons of silver: $18.5 \times 250 \text{ MWh} = 4,625 \text{ MWh}$ is roughly the equivalent of the yearly electricity consumption of 350-400 US households. Producing the aluminium requires $10,252 \text{ tons} \times 190 \text{ gigajoules per ton (GJ/ton)} = \sim 1,948,000 \text{ GJ}$, equivalent to the yearly electricity consumption of over 100,000 average households.

62 UN Trade and Development. (2025). Trade in critical minerals shapes energy transition, digital transformation and industrial development worldwide [UNCTAD SDG Pulse 2025 – In Focus]. Retrieved December 28, 2025, from <https://sdgpulse.unctad.org/critical-minerals/>.

63 Burton, M., & Attwood, J. (2025, December 23). Copper hits \$12,000 for first time as tariff trade upends market. Bloomberg News. Retrieved December 28, 2025, from <https://tinyurl.com/4jfsku9z>.

64 Mining outages are cessation of mining operations. Recently, Congo experienced fatal mining outages.

65 Bloomberg. (2025, December 24). Copper set for best year since 2009 after December surge. Bloomberg News. Retrieved December 28, 2025, from <https://tinyurl.com/yswks2dk>.

66 Attwood, J. (2025, December 10). Copper price record: How supply crunch is looming just as AI boom hits. Bloomberg News. Retrieved December 28, 2025, from <https://tinyurl.com/568jwetm>.

The world will also soon run short of copper at the current rate of rising power demand, in part due to the proliferation of AI data centres. A single example illustrates the investment and operational costs involved. It would take 1,194 truckloads to provide the 2,866 tons of copper needed for a 1 GW wind turbine—that's just for the copper.

To produce 2,866 tons of copper from ore with a 0.6 per cent yield, miners must process about 167–200 tons of ore per ton of copper. Current estimates (2020s) place the average for operating mines at approximately 0.5–0.6 per cent with many large mines below 0.6 per cent and new projects even lower (around 0.4–0.5 per cent). The formula is: Total material required (tons) = Copper produced (tons) ÷ Yield (decimal). Total material = 2,866 ÷ 0.006 = 477,666.67 tons. Divided by 400-ton trucks, this equals 1,194 truckloads.

This already assumes only the copper-bearing ore, no waste rock, no overburden,⁶⁷ no rejected material, no processing losses. In real mines, the total material moved is typically 2–4 times higher once waste rock stripping is factored in. If fully accounted for, total material moved per GW of wind power would likely exceed 1–2 million tons, not 0.48 million.

Source: Industry Sources and Own Calculations.

10.47 As demand accelerates, advanced economies are responding by promoting standards-based critical mineral markets, emphasising sustainability, traceability, and governance. Initiatives such as the G7 Roadmap to Promote Standards-Based Markets for Critical Minerals⁶⁸ aim to enhance transparency, reduce concentration risks, and encourage responsible sourcing. However, implementing digital traceability systems, meeting certification requirements, and enhancing environment, social and governance (ESG) compliance can entail substantial costs. While these objectives are legitimate, standards are not neutral technical tools. They are instruments of market power. Their design will determine who can enter supply chains, who captures value, and who bears transition costs.

10.48 From the perspective of developing countries, the current direction presents three significant challenges. Firstly, there's a risk that standards could turn into barriers instead of facilitating progress. With the introduction of digital traceability systems, certification requirements, and ESG compliance, the upfront costs and ongoing expenses can be quite steep. For many resource-rich developing nations, these financial burdens may lead to investor pessimism, and slow down project development, and limit supply. This is especially concerning as the world urgently needs these countries to ramp up their efforts in response to global transitions.

10.49 Second, there is a real concern that standards wherein if narrowly defined or asymmetrically enforced, can trap developing countries in the lowest-value segments

⁶⁷ Overburden in copper mining is the waste rock and soil removed to access the ore.

⁶⁸ G7 Canada (2025, November 01). Roadmap to promote standards-based markets for Critical Minerals. Retrieved on December 22, 2025 from: <https://tinyurl.com/3xxkck6u>.

of supply chains, exporting raw materials while concentrating value-added processing and manufacturing in advanced economies.

10.50 Third, affordability is adversely impacted. Sustainability premiums that tend to raise mineral prices without parallel support for finance, technology, and capacity-building will increase costs to transitioning globally and disproportionately impact emerging economies. A transition that is clean but unaffordable is neither rapid nor just. A durable global framework for critical minerals must therefore move beyond compliance-centric thinking. It must be inclusive, capacity-sensitive, and development-oriented. Resource-rich regions in Africa, Latin America, and Asia must be treated as co-producers of value, rather than merely sources of raw materials. This requires careful deliberation on aspects of international cooperation on technology transfer, skills, institutions, and investment in mining, processing, and recycling.

10.51 India's strategy reflects this balance with a focus on domestic capabilities through the National Critical Mineral Mission along with suitable incentive mechanism, while engaging in international partnerships like the Minerals Security Partnership⁶⁹ and the Indo-Pacific Economic Framework.⁷⁰ India consistently demonstrates that there is a complementarity between strategic autonomy and global integration.

10.52 The Government of India has launched the National Critical Mineral Mission (NCMM)⁷¹ as a strategic initiative to secure the supply chain of minerals essential for renewable energy and storage technologies. The major objectives of the Mission are to secure domestic and foreign sourcing of critical minerals and strengthen value chains.

10.53 Amendments to the Mines and Minerals (Development and Regulation) Act 2003⁷² have broadened the central government authority to auction 24 of 30 identified critical minerals and opened exploration and mining of six previously restricted minerals to private sector participation. Further, the MMDR Act was amended with effect from 01.09.2025. Through the said amendment, provision has been made in the Act for inclusion of any new mineral in a mining lease. This will boost mining of critical minerals in the country.

10.54 The Geological Survey of India has completed 195 exploration projects in FY 24-25 and taken up 230 exploration projects in FY 2025-26 for critical and strategic

69 Press Information Bureau (2023, August 07). Strengthening of Mineral Supply Chains. Retrieved on December 22, 2025 from: <https://tinyurl.com/b35vdmhh>.

70 Press Information Bureau (2024, September 22). India signs first-of-its-kind agreements focused on Clean Economy, Fair Economy, and the IPEF Overarching arrangement under Indo-Pacific Economic Framework for prosperity. Retrieved on December 22, 2025 from: <https://tinyurl.com/3btmf3b4>.

71 Press Information Bureau (2025, September 06). India's Critical Mineral Mission: Securing the Minerals of Tomorrow. Retrieved on December 23, 2025 from: <https://tinyurl.com/mwpraj8t>.

72 Press Information Bureau (2023 December 06). Launching of Critical Mineral Blocks Auction. Retrieved on December 24, 2025 from: <https://tinyurl.com/ppf88rp2>.

minerals across the country.⁷³ Meanwhile, the government's joint venture, Khanij Bidesh India Ltd. (KABIL), has acquired 15,703 hectares in Argentina for lithium mining, alongside partnerships in Australia and Chile.

10.55 To further enhance the resilience of our supply chains, the Union Cabinet has approved the setting up of the National Critical Mineral Mission (NCMM) on 29 January 2025 with a financial outlay of ₹16,300 crore and expected investment of 18,000 crore from PSUs and others. The Mission aims to secure a long-term sustainable supply of critical minerals and strengthen India's critical mineral value chains encompassing all stages from mineral exploration and mining to beneficiation, processing, and recovery from end-of-life products.

10.56 To build supply chain resilience in critical minerals, the Union Cabinet has approved a ₹1,500 crore Incentive Scheme for Promotion of Critical Mineral Recycling under NCMM. The scheme will incentivise that part of the recycling value chain in which actual extraction of critical minerals takes place, and will develop recycling capacity for critical materials in the country for the separation and production of critical minerals from secondary sources.

Carbon Credit Trading Scheme: from framework to implementation

10.57 India has made substantial progress in establishing its carbon market framework, a crucial step towards developing its mitigation strategy. The government adopted the Carbon Credit Trading Scheme (CCTS) in June 2023, operating through a dual mechanism that incorporates mandatory compliance and voluntary offset approaches.⁷⁴ The compliance mechanism targets energy-intensive industrial sectors through an emission intensity-based baseline-and-credit system, initially covering sectors such as cement, iron and steel etc.⁷⁵ Entities that exceed their emissions intensity targets earn Carbon Credit Certificates (CCCs), denominated in tonnes of CO₂ equivalent (tCO₂e), which they can trade on power exchanges. Those that fall short must buy and surrender equivalent credits. This framework leverages the existing Perform, Achieve and Trade (PAT) scheme infrastructure, gradually transitioning it into a fully operational compliance carbon market. In 2025, the government notified pro-rata Greenhouse Gas Emission Intensity (GEI) targets for four sectors: Aluminium, Cement, Chlor-Alkali, and Pulp and Paper.

10.58 Under the Offset Mechanism, Non-Obligated Entities may voluntarily register projects that reduce, remove, or avoid greenhouse gas emissions to earn CCCs. This

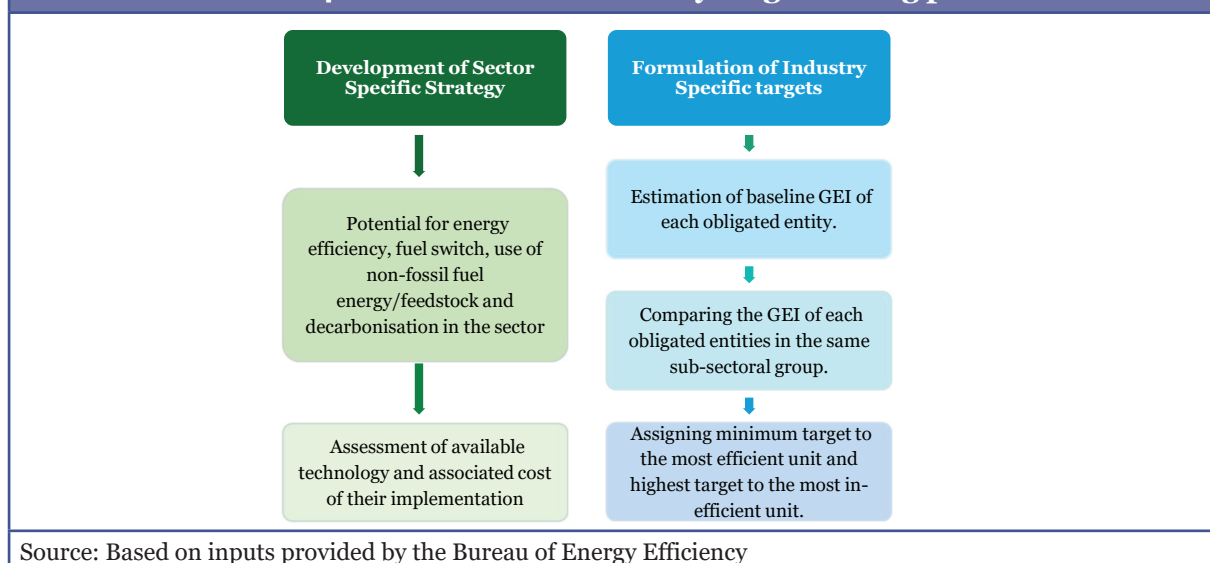
⁷³ Press Information Bureau (2024, January 15). Mineral Exploration Under Nmep. Retrieved on December 24, 2025 from: <https://tinyurl.com/3amkkes9>.

⁷⁴ Press Information Bureau (2025, December 04). Framework for Carbon Credit Trading Scheme (CCTS). Retrieved on December 10, 2025 from: <https://tinyurl.com/329s5enu>.

⁷⁵ Information Retrieved on 12 December 2025 from <https://beeindia.gov.in/carbon-market.php>.

mechanism enables mitigation outcomes from entities outside the compliance framework and incentivises climate action in these areas. In this context, the government has approved a list of 10 sectors - energy, industries, agriculture, waste handling and disposal, forestry, transport, fugitive emissions, construction, solvent use, and Carbon Capture, Utilisation, & Storage (CCUS) and others - for the offset mechanism.

Chart X.4: GHG emission intensity target setting process



Box X.5: Developments in the International Carbon Markets

Several major economies have operationalised emissions trading systems (ETS) as market-based instruments to reduce greenhouse gas emissions, offering useful lessons for countries designing or scaling domestic carbon markets. The European Union Emissions Trading System, the Korea Emissions Trading Scheme, and the China National ETS represent three of the world's major emissions trading systems, reflecting different design choices and stages of market maturity.

The EU ETS, launched in 2005, is a cap-and-trade, compliance-based system with an absolute emissions cap that is progressively tightened over time. Evidence suggests that in the initial stages of the EU ETS, oversupply of offset credits led to dilution of emission prices. The ETS, now in its fourth phase, is highly complex, encompassing power, industry, domestic aviation, and maritime sectors in its current form. In general, the allowances accounting for up to 57 per cent of the emission cap are to be auctioned, and the rest are allocated freely to avoid “carbon leakage”⁷⁶ in susceptible sectors, such as steel, cement, aluminium, fertilisers, hydrogen, and iron and steel. The Carbon Border Adjustment Mechanism (CBAM) will gradually replace free allocation in such sectors from 2026 onwards.

The EU has further introduced a new emissions trading system (ETS2) to cover CO₂ emissions from fuel combustion in buildings, road transport, and additional sectors (mainly

⁷⁶ EU states that ‘Carbon leakage could occur if ETS-regulated activities were moved to non-EU countries with less ambitious climate policies, leading to an increase in overall greenhouse gas emissions.’ in European Union (2025, December 03). Retrieved on December 23, 2025 from: <https://tinyurl.com/mp5rb37d>.

small industries not covered by the existing EU ETS) (to be implemented by 2027).⁷⁷ The use of offset credits, initially allowed, has now been discontinued in Phase Four.

Korea's ETS,⁷⁸ operational since 2015, follows a similar cap-and-trade approach but places greater emphasis on calibrated free allocation and limits the use of offsets to manage competitiveness concerns. Korean ETS covers the maritime, waste, domestic aviation, transport, buildings, industry, and power sectors, with free allocations of allowances based on a benchmark system, which allows entities to use offsets up to 5 per cent of their verified emissions.

China's national ETS,⁷⁹ implemented in 2017, covers the power, steel, cement, and aluminium smelter sectors, with annual emissions exceeding 26,000 tCO₂, and is an intensity-based system. The allowances in the Chinese ETS are 100 per cent freely allocated using an output-based (or-benchmark)⁸⁰ approach, while also allowing entities to use offsets up to five per cent of their verified emissions.

A few lessons from these experiences in the context of the carbon markets framework are (a) a gradual approach to inclusion of sectors, (b) avoiding generous free allowances or allowing offsets to meet the targets and (c) ensuring alignment with country contexts.

Mission LiFE

10.59 The Mission LiFE - Lifestyle for Environment, an initiative introduced in 2021, at COP26 in Glasgow, connects individual and community behaviour change with efforts to deal with climate change. In fact, LiFE is an integral part of India's NDC.

10.60 By driving core shifts in consumption and production patterns towards sustainability, embedding circularity, and enabling adaptation to the impacts of climate change, the Mission's functions highlight that small, collective actions can cumulatively advance the country's climate goals.

10.61 Majority of India's climate-oriented schemes are fundamentally aligned with the ethos of Mission LiFE, as they combine government interventions with behavioural and lifestyle shifts at the household, community and enterprise levels. In energy efficiency, programmes such as appliance Standards and Labelling, mass LED adoption (Unnat Jyoti by Affordable LEDs for All -UJALA), and industrial efficiency

⁷⁷ European Union. Retrieved on December 23, 2025 from: <https://tinyurl.com/3jx8jkpx>.

⁷⁸ International Carbon Action Partnership. Korea Emissions Trading System (K-ETS). Retrieved on December 23, 2025 from <https://tinyurl.com/4akssbd8>.

⁷⁹ International Carbon Action Partnership. China National ETS. Retrieved on December 23, 2025 from <https://icapcarbonaction.com/en/ets/china-national-ets>.

⁸⁰ 'The sectoral balance value represents the carbon emissions intensity level at which the total number of allowances allocated would equal total verified emissions' in International Carbon Action Partnership (2025, December 01). China releases 2024–2025 allowance allocation plan for industrial sectors in National ETS. Retrieved on December 23, 2025 from <https://tinyurl.com/5buz49xd>.

mechanisms (PAT) encourage the use of less energy for the same service, embedding mindful consumption rather than only supply-side control on emissions. The NAPCC provides the overarching framework for India's climate response, with its missions on energy efficiency, sustainable habitats, water, and agriculture explicitly promoting conservation, resource efficiency, and community participation, which are also the core principles of Mission LiFE.

10.62 Together, these examples demonstrate that India's climate strategy is not confined to emissions targets or technologies alone, but is deliberately designed to reshape consumption patterns, social norms and daily choices, making Mission LiFE not a parallel initiative but the behavioural foundation underlying most climate policies in the country.

Box X.6: Transforming Waste into Wealth by Odisha⁸¹

Odisha launched a state-wide mission to tackle solid waste in 2023, called the “Garbage-Free Urban Odisha”. The mission combines advanced bio-mining technologies, strong institutional planning, and community participation through self-help groups to tackle legacy waste and transform reclaimed dumpsites into new wealth centres, thereby securing against further dumping. The Bhubaneswar Municipal Corporation has rolled out initiatives such as a coconut waste processing facility, a flower waste facility, a 150-tonnes per day (TPD) mega material recovery facility, and a 10-TPD plasma processor to convert waste into value-added products, creating livelihoods, and fostering innovation in urban waste management.

Climate Finance

10.63 The current levels of climate finance fall short of the requirements of developing countries to meet their climate ambitions. By 2030, developing economies are estimated to need USD 5–6 trillion⁸² for climate action. The Fourth International Conference on Financing for Development, titled *Compromiso de Sevilla*⁸³ recognised that, despite sustained global efforts, the gap between sustainable development ambitions and available financing has continued to widen—particularly for developing countries—reaching an estimated USD 4 trillion annually. This gap does not arise from a lack of global capital, but rather from the disparity between where capital is concentrated and where investment needs for sustainable development are most pressing.

⁸¹ Inputs provided by Government of Odisha.

⁸² UNFCCC (2024, September 10). Second report on the determination of the needs of developing country Parties related to implementing the Convention and the Paris Agreement. Retrieved on December 23, 2025 from: <https://unfccc.int/documents/640757>.

⁸³ UNDESA. Sevilla Commitment Fourth International Conference on Financing for Development. Retrieved on December 23, 2025 from: <https://tinyurl.com/mrxk22e9>.

10.64 Global financial assets under management reached USD 147 trillion in 2025,⁸⁴ yet climate finance flows remain highly skewed. In 2023, global climate finance totalled USD 1.9 trillion, of which private capital accounted for nearly USD 1.3 trillion. More than half of this private finance flowed to advanced economies, with China attracting another 30 per cent, while developing countries, excluding China, received only about 15 per cent.⁸⁵ International public finance to developing economies remains limited, and domestic actors continue to dominate global climate finance, accounting for nearly 80 per cent of total flows.

10.65 These patterns embedded in the international financial architecture reflect a persistent and clear bias in favour of developed countries. Capital tends to flow to the economies with deeper financial markets, stable macroeconomic conditions, and those that are generally perceived as having lower risks. Developing countries, however, face higher borrowing costs because of currency volatility, lower sovereign credit ratings, and financial systems that lack depth. As a result, despite the abundant global capital, investors often hesitate to finance climate resilience projects in developing countries, even when these nations are more vulnerable to climate change impacts and such investments could yield significant benefits. Investors, do not internalise the social benefit of their investments in developing countries [Box III.3: Microfinance, financialisation, and the need to re-centre household welfare in impact investing]. This imbalance not only hampers growth and development in these countries but also undermines global efforts to address the escalating climate crisis.

10.66 India faces global challenges in climate finance. Although the country has successfully reduced its emissions intensity by 36 per cent since 2005 and achieved 50 per cent non-fossil power capacity ahead of schedule,⁸⁶ climate finance remains skewed towards mature sectors such as solar, wind energy and energy efficiency. Critical areas, including adaptation, financing for micro, small, and medium enterprises (MSMEs), urban infrastructure, and hard-to-abate industries, remain underfunded. Currently, around 83 per cent of India's finance for mitigation and 98 per cent of finance for adaptation is sourced domestically.⁸⁷ However, the gaps in available finance and the needs persist,⁸⁸ relying solely on domestic resources will not be sufficient.

84 Torbey, H., Kwek, J.-H., Banani, F., & Nguyen, V. (2025, September 18). Asset management 2025: The great convergence. McKinsey. Retrieved December 24, 2025, from: <https://tinyurl.com/372k2338>.

85 Climate Policy Initiative. 2025. Global Landscape of Climate Finance 2025. Retrieved December 24, 2025, from <https://tinyurl.com/bde426bf>.

86 Press Information Bureau. (2025, December 08). Parliament Question: Reduction In Carbon Dioxide Emission. Retrieved on December 23, 2025 from: <https://tinyurl.com/vhtmkfv>.

87 Climate Policy Initiative. 2024. Landscape of Green Finance in India 2024. Retrieved December 24, 2025, from <https://www.climatepolicyinitiative.org/publication/landscape-of-green-finance-in-india-2024/>

88 Climate Policy Initiative. 2024. Landscape of Green Finance in India 2024. Retrieved December 24, 2025, from <https://www.climatepolicyinitiative.org/publication/landscape-of-green-finance-in-india-2024/>.

10.67 International public sector climate finance at an affordable cost, is therefore essential for mobilising private sector finance, required to meet climate ambitions.

Bridging the Finance Gap in the Indian Context

10.68 India has taken several steps to mobilise climate finance. Sovereign green bonds have been issued to fund low-carbon public infrastructure, providing policy signalling and market benchmarks. The broader ecosystem has also been strengthened. 100 per cent FDI has been permitted under the automatic route for renewable energy projects.⁸⁹ Development finance institutions, such as Indian Renewable Energy Development Agency Ltd (IREDA),⁹⁰ National Bank for Agriculture and Rural Development (NABARD),⁹¹ Small Industries Development Bank of India (SIDBI),⁹² Power Finance Corporation Ltd. (PFC),⁹³ and Rural Electrification Corporation Ltd. (REC),⁹⁴ offer credit lines and financing schemes for climate-related investments. SEBI's Business Responsibility and Sustainability Reporting (BRSR) framework, green bond guidelines and IFSCA's guidance on sustainability-linked lending have improved disclosure quality and investor confidence in climate-related investments. RBI has also introduced the green deposit framework that optimises the flow of credit to green activities/projects by channelising institutional and household savings, with guardrails in place to overcome greenwashing challenges. Chart X.5 provides an overview of the major financial regulations introduced in recent years in this context.

10.69. The financing of disaster risks in India has also evolved beyond response and relief (through the State/National Disaster Response Fund) to incorporate risk mitigation, reconstruction, and recovery, as well as prevention, under the State Disaster Mitigation Fund (SDMF)⁹⁵ and the National Disaster Mitigation Fund (NDMF),⁹⁶ institutionalised as part of the Disaster Management Act 2005. These funds aim to reduce the risks, impact, or effects of disasters through project-based funding, encouraging local-level and community-based interventions. The Glacial Lake Outburst Flood Mitigation

89 19 DEC 2023 India's renewable energy sector has received FDI equity investment of \$ 6.1 billion during April 2020 – September 2023: Union Minister for Power and New & Renewable Energy. <https://www.pib.gov.in/PressReleaseIframePage.aspx?PRID=1988293®=3&lang=2>

90 <https://www.ireda.in/schemes>

91 <https://www.nabard.org/about-departments.aspx?id=5&cid=2788>.

92 <https://www.sidbi.in/uploads/financialreport/Annual-Report-for-FY-2024-25-Part-1.pdf>

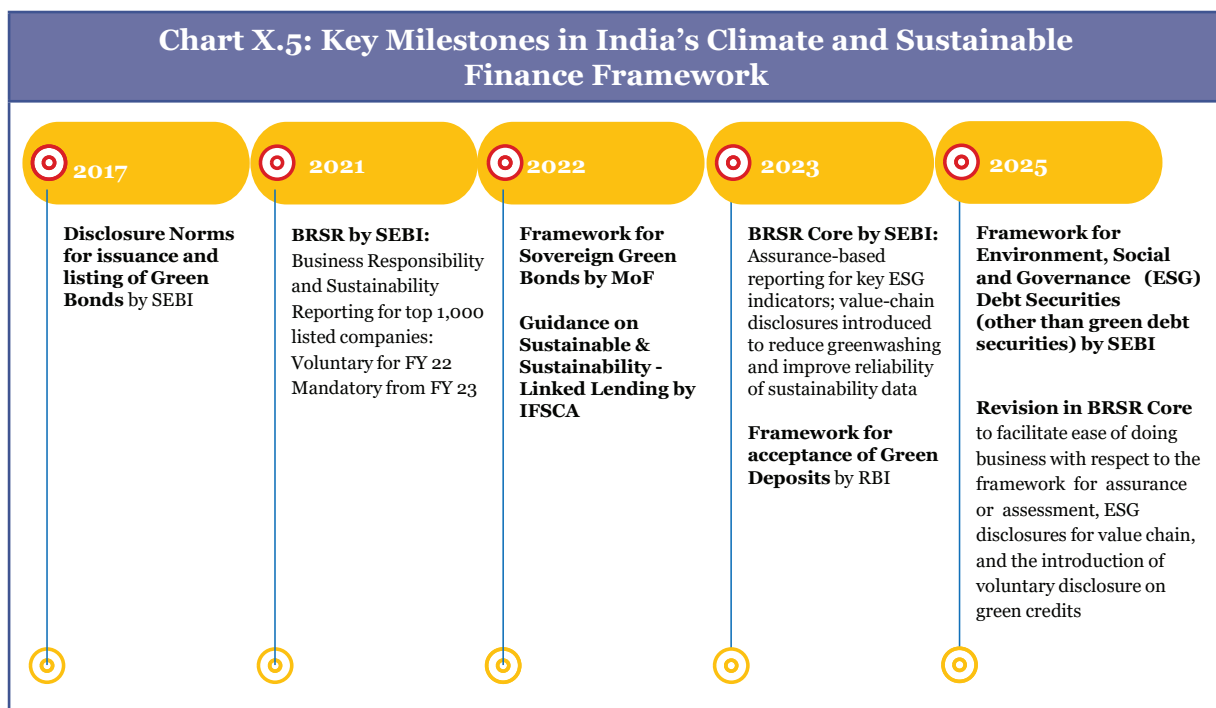
93 https://www.pfcindia.co.in/ensite/DocumentRepository/ckfinder/files/Investors/ESG_Report/ESG%20Standalone%20report%20FY%202024-25.pdf.

94 <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2002548®=3&lang=2>.

95 Ministry of Home Affairs, Government of India. (2022, January 14). Guidelines on constitution and administration of the State Disaster Mitigation Fund (SDMF) based on the recommendations of the Fifteenth Finance Commission 2021–22 to 2025–26 [PDF]. Retrieved December 28, 2025, from <https://tinyurl.com/49xk678v>.

96 Ministry of Home Affairs, Government of India. (2022, February 28). Guidelines on constitution and administration of the National Disaster Mitigation Fund (NDMF) based on the recommendations of the Fifteenth Finance Commission 2021–22 to 2025–26 [PDF]. Retrieved December 28, 2025, from <https://tinyurl.com/hswd7jr6>.

Programme has been recently approved under NDMF to monitor glaciers and glacial lakes in the Indian Himalayan region.⁹⁷



10.70 Despite these measures, challenges persist in accessing finance at scale in India. The cost of capital for climate projects remains high, access to multilateral finance is often complex, and long-term international capital from institutional investors remains limited. Risk-mitigation and risk-sharing mechanisms are still underdeveloped, particularly for emerging technologies and large-scale adaptation projects.

10.71 The following section discusses a two-pronged strategy for scaling up finance for climate action from both domestic and international sources.

Strengthening the Domestic Financial System

Deep and Liquid Bond Markets

10.72 Bond markets are crucial for financing climate infrastructure, which requires substantial upfront capital and extended repayment horizons. Deeper, more liquid bond markets can provide long-term, stable, and scalable financing at predictable costs. On one hand, mature markets are important for attracting investments from institutional investors, which have long-term capital at their disposal. On the other hand, bond markets provide an important platform for local administrative bodies to raise local-currency finance for climate-aligned functions, such as water supply, waste

⁹⁷ Press Information Bureau. (2025, February 4). GLOF mitigation project. Retrieved December 28, 2025, from <https://tinyurl.com/msfhz4kr>.

management, and green energy, tailored to area-specific adaptation and resilience needs. Urban local bodies in Indore, Ghaziabad, Ahmedabad, and Vadodara have issued green bonds in line with SEBI's green bond framework (Table X.I). Municipal green bonds can unlock USD 2.5–6.9 billion for local bodies driven climate action over the next 5–10 years.⁹⁸ Besides, government of India has now issued sovereign green bonds worth ₹15,000 crore in FY26, with the cumulative issuance reaching ₹72,697 crore since FY23 (ChartX.6).⁹⁹

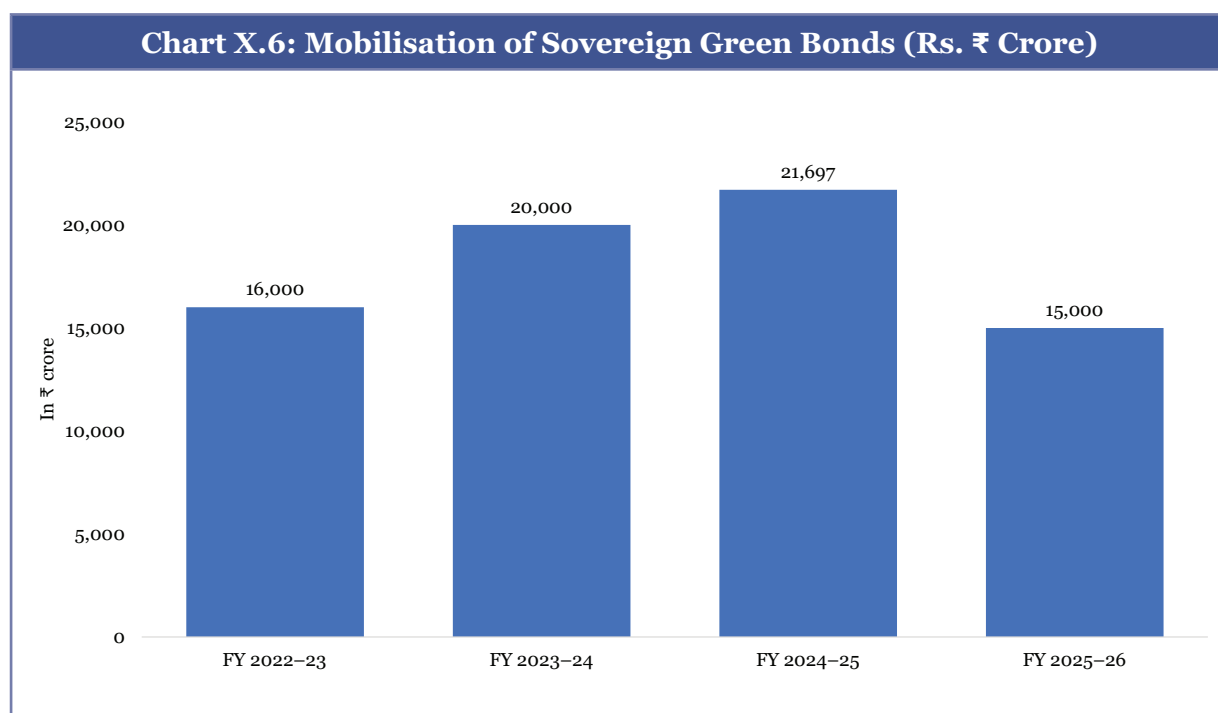


Table X.1: Municipal Green Bonds issues in India

Municipal corporation	Date of issuance	Amount Raised (₹crore)	Use of proceeds	Tenure
Vadodara	Feb-24	100	Water treatment	5
Ahmedabad	Feb-24	200	Water treatment	5
Indore	Feb-23	244	Solar plant	3 to 9
Ghaziabad	May-25	150	Sewage Treatment	4.02 to 10.02

Source: ESG Debt Securities, SEBI. Accessed on 27 December 2025 from: <https://www.sebi.gov.in/statistics/greenbonds.html>.

⁹⁸ Bibhudatta, A., & Rathee, D. (2025). Unlocking Green Finance for India's Urban Local Bodies through Municipal Green Bonds.

⁹⁹ Date provided by Public Debt Management Agency, DEA.

Box X.7: Greenium in Sovereign Green Bond Markets — Cross-Country Comparison and Market Issues¹⁰⁰

Greenium—the yield advantage of green bonds over comparable conventional bonds—has been observed across several sovereign issuers, but its magnitude and persistence vary significantly by market. Cross-country experience shows that greenium outcomes depend less on investor intent alone and more on market design, liquidity, credibility, and reporting frameworks.

Key Market Issues Affecting Greenium across Countries

Country	Greenium	Key Market Strengths	Key Constraints
European Union	~2 bps, stable	Large benchmark-sized issuances; strong supranational credit quality; robust allocation & impact reporting; deep investor base	Limited scope for higher Greenium due to already low yields and very high liquidity in conventional EU debt.
United Kingdom	Small but positive, episodic	Integration with gilt programme; focused issuance strategy; strong institutional investor participation	Initial fragmentation across maturities; Greenium sensitive to overall gilt market volatility
France	~2–5 bps, consistent	Early mover advantage; large issue sizes; strong ESG investor base; detailed impact reporting	Complexity of reporting across many eligible sectors increases monitoring costs
India	Intermittent, 0–6 bps	Clear sovereign green bond framework; strong domestic institutional demand; policy signalling value	Limited secondary market liquidity; smaller issue sizes; fragmented issuance; lagged impact reporting; green bonds largely held to maturity
Emerging Markets (general)	Weak or absent	High climate investment needs; growing ESG interest	Higher sovereign risk premiums; currency risk; shallow bond markets; weak benchmark comparability

International experience shows that greenium is not automatic and tends to be durable only when sovereign green bonds are issued in large, liquid benchmark sizes, allow clear comparability with conventional bonds, provide credible and timely allocation and impact reporting, and are embedded within a stable sovereign debt programme.

From the foregoing, it could be assessed that, in India, the modest and episodic greenium observed so far does not necessarily reflect a lack of investor interest. Deepening secondary

¹⁰⁰ RBI-NIPFP study for DEA, Factors determining the presence of Greenium in Sovereign Green bonds

market liquidity, consolidating issuances into benchmark sizes, and strengthening post-issuance reporting can help translate into a more significant cost-of-capital advantage over time.

10.73 Strengthening National development finance institutions – Specialised financial institutions such as IREDA, NABARD, SIDBI, PFC, and REC are already working in the low-carbon/renewable energy space, promoting the adoption of sustainability practices and encouraging green investments through key initiatives and schemes. These institutions support climate project preparation and augment the bankability of projects through catalytic capital, which closely intersect with India's development priorities including climate action. Further strengthening of the financial ecosystem through the mobilisation of blended finance, de-risking of projects, and capacity building through technical assistance and training are vital for enhancing the flow of resources for climate action.

10.74 Availability of insurance cover against climate risks – Expanding insurance coverage is essential to safeguard people against economic losses associated with the physical risks of climate change, but it also improves the creditworthiness of climate-exposed borrowers such as farmers and MSMEs. In India, the role of insurance is particularly significant for the agriculture sector, and is discussed in greater detail in Chapter VI Agriculture and Food Management.

International Climate Finance and the Role of Multilateral Development Banks

10.75 International climate finance is critical for the global response to climate change. Article 9, subparagraph 1 of the Paris Agreement clearly mandates developed countries to provide financial resources to developing countries for their climate action, in line with equity and the principle of common but differentiated responsibilities and respective capabilities, in light of national circumstances. With respect to international commitments on climate finance – the USD 100 billion annual goal till 2025, while developed countries reported providing USD 115.9 billion in climate finance to developing countries in 2022,¹⁰¹ these figures are widely contested, with contrasting estimates suggesting that actual provision may be only USD 28–35 billion.¹⁰² With the continuing lack of clarity on climate finance provision under the Paris Agreement, COP 30 in November 2025, in Belem, Brazil, arrived at an important decision on Article

¹⁰¹ OECD (2024), Climate Finance Provided and Mobilised by Developed Countries in 2013–2022, Climate Finance and the USD 100 Billion Goal, OECD Publishing, Paris, <https://doi.org/10.1787/19150727-en>. Retrieved on 27 December 2025 from <https://tinyurl.com/49u3j6je>.

¹⁰² Oxfam & CARE. (2025). Climate Finance Shadow Report 2025: Analysing progress on climate finance under the Paris Agreement. CARE Climate Change (CARE) & Oxfam Policy & Practice (Oxfam). Retrieved on 27 December 2025 from <https://tinyurl.com/5n77trm5>.

9.1 of the Paris Agreement, establishing a two-year work programme under the COP.¹⁰³ This outcome helps shift the focus back to obligations rather than private investments being counted as climate finance.

10.76 The disparity between climate vulnerability and financial capacity among nations needs to be acknowledged and addressed to ensure an effective global response to climate change, which is currently limited by both the insufficient volume of finance and high capital costs for developing countries. These costs are shaped by factors such as credit ratings, risk perceptions, and the structure of the global financial system.

Box X.8: Sustainable and Climate Finance — Plenty of Capital, Limited Risk Appetite¹⁰⁴

Despite sustained global efforts, financing for climate action remains far below requirements, particularly in developing countries. The impacts are evident in persistent energy poverty and rising climate vulnerability. The core challenge is not a shortage of global capital, but a structural misalignment between abundant liquidity and risk appetite.

Global capital markets are flush with funds, yet flows to sustainable development and climate projects in the Global South remain constrained by entrenched risk aversion embedded in the architecture of global finance. This is most evident in the operating models of Multilateral Development Banks (MDBs) and in the prudential regulations of developed countries. MDBs continue to prioritise low-risk, sovereign-backed lending and the preservation of AAA ratings, limiting balance-sheet recycling and private capital mobilisation. A shift toward balance-sheet optimisation—from “originate-to-hold” to “originate-to-share”—is essential to reposition MDBs as global risk managers, utilising guarantees, insurance, and blended finance to attract private investment.

These constraints are reinforced by regulatory frameworks such as Basel III and Solvency II, which impose high capital charges on long-term, MDB-backed infrastructure investments in emerging economies, effectively excluding large pools of private capital. Further, an asset-class mismatch persists: institutional investors prefer standardised, liquid securities, while developing-country projects are bespoke and fragmented.

Unlocking scale requires MDB recapitalisation, strategic use of public capital as first-loss finance, regulatory recalibration, and strong governance safeguards. Reforms across MDB mandates, global regulations, and domestic financial systems are critical to align climate action with development priorities and deliver inclusive, resilient growth.

¹⁰³ Press Information Bureau. (2025, November 23). India welcomes key outcomes at UNFCCC COP30; reaffirms commitment to equity, climate justice and global solidarity [Press release]. Government of India. Retrieved December 28, 2025, from <https://tinyurl.com/4nb5k2c8> and Third World Network. (2025, November 24). COP30 ends with delicate Belém Political Package amid developing country concerns. Retrieved December 28, 2025, from <https://tinyurl.com/3kejvyr9>.

¹⁰⁴ CEA address ASPIRE to Scale Reimagining Development Finance: Blended and Innovative Finance Tools for Asia's Economic Growth - Date: 14 October 2025

Recent innovations demonstrate how targeted risk-sharing can unlock private capital for climate action. In 2025, the Inter-American Development Bank and the Central Bank of Brazil¹⁰⁵ established a mechanism to unlock up to USD 3.4 billion in long-term foreign-exchange hedging, directly addressing a key deterrent to investment in emerging markets. By mitigating currency risk without adding to sovereign debt, the arrangement enhances project bankability and demonstrates how MDB balance-sheet optimisation can attract private finance. Such approaches highlight the need for a broader shift from risk avoidance to active risk management in global climate finance.

10.77 These outcomes highlight structural weaknesses in the international financial system. Current global capital allocation often overlooks the long-term benefits of investing in climate solutions, as well as the role of public policy and international cooperation. As a result, developing countries must manage their climate action-related investments with shorter-term loans, higher risk premiums, and a greater need to utilise their own resources, which limits their ability to implement effective climate action.

10.78 Developing countries' climate efforts will depend on a range of systemic reforms that must be coordinated across MDB mandates, global regulations, blended finance tools, and domestic financial systems. These changes can help align climate action with development goals, support economic resilience, and encourage inclusive growth. However, it is important that such adjustments do not divert capital away from other development priorities.

10.79 Blended finance tools, particularly those involving government support, should be utilised with caution. While such instruments help in attracting private investment, they also create fiscal trade-offs for governments. Using public sector funds to de-risk climate projects for private investors can create liabilities and put more pressure on the limited fiscal space if the risks materialise. Strategies for scaling finance for climate action should strike a balance between attracting investment and the need for fiscal responsibility, transparency, and long-term debt sustainability. In that regard, it is to be mindful of the fact that excessive use of foreign currency borrowing also exposes projects to exchange rate risks, increasing forex premiums and financing costs, ultimately impacting consistency in climate action. Strong governance frameworks are necessary to ensure that climate finance supports growth and resilience without compromising macroeconomic stability.

¹⁰⁵ Organisation for Economic Co-operation and Development. (2024). Case study – Foreign private capital mobilisation and currency hedging: Inter-American Development Bank and Brazil (OECD Case Studies). Retrieved December 28, 2025, from <https://tinyurl.com/t97m4dd3>.

Regulation: Streamlining environmental governance for growth

10.80 Environmental regulation finds its theoretical basis in welfare economics, recognising pollution as a negative externality that leads to market failure if left unchecked. Environmental regulation assumes a central role, not merely to correct these market failures, but to enable sustainable growth.

10.81 Traditionally, environmental regulation has been guided by Pigouvian welfare economics,¹⁰⁶ command-and-control approaches, which prescribe emission standards or technology norms that provide certainty but often impose high compliance costs and weak incentives for innovations.¹⁰⁷ This approach often entails higher compliance costs and may weaken incentives for innovation. In contrast, market-based instruments, such as pollution taxes, tradable permits, and emissions trading, offer greater flexibility and cost-efficiency. More recent regulatory theories,¹⁰⁸ including the toolkit prepared by OECD,¹⁰⁹ emphasise risk and outcome-based regulation, calibrating stringency to environmental risk and assessing compliance by results rather than processes. Thus, well-designed regulations can reduce unnecessary burdens, promote innovation, and reconcile environmental protection with ease of doing business.¹¹⁰

Evolution of Environmental Regulation in India

10.82 Environmental policymaking in India gained momentum in the aftermath of the 1972 Stockholm Declaration. This led to the enactment of the Water (Prevention and Control of Pollution) Act, 1974, which laid the institutional foundation for pollution control by establishing State-level Pollution Control Boards (SPCBs) to prevent and control water pollution. These Boards were vested with the authority to initiate suo-motu action against violations of environmental laws. To complement this framework, the Water (Prevention and Control of Pollution) Cess Act, 1977, was introduced, enabling the levy of a cess on polluting and water-intensive industries based on their water consumption.

10.83 The Bhopal Gas Tragedy of 1984 marked a critical turning point in India's environmental governance with the enactment of the Environment (Protection) Act, 1986 (EPA). The EPA functions as an umbrella legislation that confers powers on the central government to regulate environmental pollution, enforce compliance, and issue subordinate rules and notifications. The three core statutes, i.e., the Water Act, the Air

106 Pigou, A. C. (1951). Some Aspects of Welfare Economics. *American Economic Review*, Vol. 41, No. 2 (Papers and Proceedings of the Sixty-Third Annual Meeting of the American Economic Association), pp. 287–302.

107 Baumol, W. J., & Oates, W. E. (1988). *The theory of environmental policy*. Cambridge university press.

108 Gunningham, N., Grabosky, P., & Sinclair, D. (1998). *Smart regulation: Designing environmental policy*. Oxford University Press,.

109 OECD (2014) *Regulatory Enforcement and Inspections toolkits*.

110 Porter, Michael E. (1991). America's Green Strategy. *Scientific American*, Vol. 264, No. 4, pp. 168.

Act, and the EPA, established the command and control framework for environmental regulation in India. In addition, the introduction of the Environmental Impact Assessment (EIA) Notification, 1994, and its replacement by the EIA Notification, 2006, institutionalised ex-ante environmental regulation.¹¹¹ The EIA regime requires prior environmental clearance for specified categories of projects, incorporating screening, scoping, public consultation, and expert appraisal. Judicial intervention has played a significant role in shaping environmental regulation in India. The establishment of the National Green Tribunal (NGT) in 2010 strengthened enforcement but also increased regulatory scrutiny and litigation risks

10.84 Over time, in India, more flexible, incentive-based mechanisms have been introduced. At the national level, the Perform, Achieve and Trade (PAT) scheme, which was launched in 2012, introduced a baseline-and-credit system that creates tradable Energy Saving Certificates (ESCs) for industrial units that outperform their energy efficiency targets, incentivising innovation and cost-effective efficiency improvements rather than just imposing fixed standards. Recently, the framework for the Indian carbon market under the CCTS has been formulated. Yet another is the ETS for particulate matter, the first of its kind in the world, launched in Surat, Gujarat (Box X.9).

Box X.9: Do Pollution Trading Schemes Work in Developing Countries?

In the United States and the European Union, pollution markets have been widely praised for translating sound economic theory into effective environmental policy. However, progress has been limited in developing countries. This is due to the assumption that market-based environmental regulations are rare in developing economies, and state capacity is also weak. In addition, regulators in developing countries often struggle to monitor emissions effectively and may lack the credibility or enforcement power required to ensure that polluters hold permits for every unit of emissions.

However, the seminal work by Greenstone et al (2023)¹¹² sets aside this argument and provides the first experimental evaluation of a particulate matter emissions trading market, the first of its kind globally implemented in a large Indian city of Gujarat (Surat). The scheme covered 317 industrial plants and was supported by the mandatory installation of Continuous Emissions Monitoring Systems (CEMS) for large particulate sources. The programme replaced a conventional command-and-control framework based on technology requirements and concentration standards. The findings brought out that (a) the market functioned effectively, with active permit trading and near universal compliance, (b) plants participating in the emissions market reduced particulate emissions by 20-30 per cent compared to plants regulated under the status quo regime and (c) for a given level of

¹¹¹ Turaga, R. M. R., & Sugathan, A. (2020). Environmental regulations in India. *Oxford research encyclopaedia of environmental science*.

¹¹² Greenstone, M., Pande, R., Ryan, N., & Sudarshan, A. (2025). Can pollution markets work in developing countries? Experimental evidence from India. *The Quarterly Journal of Economics*, 140(2), 1003-1060.

emissions, the market reduced abatement costs by 11–14 per cent, reflecting differences in firms' marginal abatement costs.

This suggests that when supported by credible monitoring infrastructure, pollution markets can deliver significant emissions reductions at lower compliance costs even in lower-capacity settings.

Recent reforms in Environmental Clearance and Related Regulations¹¹³ include:

10.85 Rationalisation and predictability in regulations have been a specific area of concern for the government. India's environmental regulatory framework is undergoing a gradual shift towards a governance model that involves expanding risk-based and outcome-oriented regulations, increasing the use of market-based instruments, strengthening the capacity and accountability of SPCBs, and enhancing regulatory predictability. Even so, several critical challenges remain, including the multiplicity of clearances and overlapping jurisdictions, weak post-clearance monitoring, capacity constraints within SPCB, significant inter-state variation in implementation, and high litigation-related uncertainty.

10.86 The Government of India has undertaken numerous measures to reform environmental governance, aligning environmental protection, climate commitments, and ease of doing business. These reforms focus on simplifying approvals, strengthening compliance, leveraging digital technologies, and transitioning to a trust-based, outcome-oriented regulatory framework.

10.87 At the centre of this transformation is PARIVESH (Pro-Active and Responsive facilitation by Interactive, Virtuous, and Environmental Single Window Hub) 3.0, a nationwide single-window digital platform for environmental clearances and post-approval compliance monitoring. The system integrates baseline environmental data, compensatory afforestation land banks, inter-ministerial dashboards, and AI-enabled 24×7 support, significantly improving transparency, predictability, and administrative efficiency.

10.88 Compliance oversight has been strengthened through the Environment Audit Rules, 2025, which have introduced certified third-party environmental auditors under major environmental statutes. This reform enhances credibility while reducing procedural burden on regulators. In parallel, following amendments of the Water Act and the Air Act, the Government of India has issued uniform guidelines for the grant of consent to establish and operate for industries across all SPCBs in the country.

¹¹³ Inputs provided by MoEFCC.

10.89 Environmental regulations have been rationalised based on pollution potential. Industry classifications have been updated to Red, Orange, Green, Blue, and White¹¹⁴ categories, to promote transition towards better compliance. Compressed Bio-gas (CBG) plants (excluding those using industrial/process waste) are classified as Blue; those generating by-products using electricity fall under White.

10.90 Given the strategic importance of critical minerals and to support development in line with climate priorities of India, regulations related to mining projects of Critical, Strategic, and Atomic Minerals have been streamlined.¹¹⁵

10.91 Ecosystem-based approaches and restoration are being incentivised through the Afforestation and Green Credit Programme that encourages public and private participation in compensatory afforestation and degraded forest restoration. Circular economy goals are being advanced through Circular Economy Action Plans covering 10 waste categories¹¹⁶ and Extended Producer Responsibility (EPR) frameworks for multiple waste streams,¹¹⁷ supported by centralised digital portals. As on 14 November 2025, total 69,116 producers and 4,377 recyclers have been registered on the portal. Approximately 308 lakh tonnes of waste (plastic packaging waste, battery waste, e-waste, waste tyres) has been recycled with corresponding EPR certificates generation of 296.53 lakh tonnes, out of which 236.93 tonnes has been transferred to producers.

10.92 To address plastic pollution, 12 identified single-use plastics (SUPs)¹¹⁸ having high littering potential and low utility banned from 1 July 2022. National and state level task forces have been formed to eliminate identified SUPs and for effective implementation of the Plastic Waste Management Rules. For effective monitoring of ban on identified SUP items and plastic waste management in the country, an online platforms (a) National Dashboard for monitoring of comprehensive action plan implementation, (b) Central Pollution Control Board (CPCB) Monitoring Module for Compliance on Elimination of

¹¹⁴ The category of the sector is decided based on the following ranges of Pollution Index:

- i. Red: $PI \geq 80$,
- ii. Orange: $55 \leq PI < 80$,
- iii. Green: $25 \leq PI < 55$,
- iv. White: $PI < 25$; and

Industries providing essential environmental services for managing domestic waste are classified as 'Blue' with an extended Consent validity of 2 years.

¹¹⁵ MOEFCC. Information retrieved on 14 January 2026, from <https://tinyurl.com/5n6z88ku>.

¹¹⁶ Lithium-ion batteries; E-waste; Toxic and hazardous industrial waste; Scrap metal (ferrous and non-ferrous); Tyre and Rubber; End of Life Vehicles; Gypsum, Used Oil, Solar Panels and Municipal Solid Waste.

¹¹⁷ plastic packaging waste, battery waste, e-waste, waste tyres, used oil, end-of-life vehicles, construction and demolition waste, and scrap of non-ferrous metals.

¹¹⁸ ear buds with plastic sticks, plastic sticks for balloons, plastic flags, candy sticks, ice-cream sticks, polystyrene [Thermocol] for decoration; Plates, cups, glasses, cutlery such as forks, spoons, knives, straw, trays, wrapping or packing films around sweet boxes, invitation cards, and cigarette packets, plastic or PVC banners less than 100 micron, stirrers.

SUP and (c) CPCB Grievance Redressal App are in operation. Additionally, the thickness of plastic carry bags was increased to 120 microns making them easier to recycle.

10.93 The Environment Protection (Management of Contaminated Sites) Rules, 2025, establish a framework for identifying and remediating contaminated sites. Biological Diversity (Access to Biological Resources and Knowledge Associated thereto and Fair and Equitable Sharing of Benefits) Regulations, 2025 aim to regulate access to India's biological resources and associated traditional knowledge including for digital sequence information.

10.94 The Public Liability Insurance Act amended through the Jan Vishwas Act decriminalises offences through penalty in place of imprisonment and relief/compensation for deaths/fatalities in case of accidents was increased from ₹ 25,000/- to ₹ 5 lakh. It allows utilisation of the Environmental Relief Fund of ₹ 1,100 crore for the remediation of contaminated sites. The rates of relief were shifted from the Act to the rules to ensure greater flexibility. Through this act, the Ministry has decriminalised "the Environment (Protection) Act, 1986, the Air (Prevention and Control of Pollution) Act, 1981 and the Indian Forest Act, 1927. Apart from it, "Water" being a subject falling under the "State List", the criminal Provisions of the Water (Prevention and Control of Pollution) Act, 1974 have been amended under Article 252(1) of the Constitution of India. All the above four Acts have been decriminalised and have rationalised minor offences to further enhance trust-based governance for ease of living and doing business.

10.95 In October 2025,¹¹⁹ green belt requirements for industries were rationalised based on the pollution potential of industrial estates/parks and individual industries, eliminating previous norms that required an overall green cover of 33 per cent of the industrial estates. The new norms require more polluting industries to have a greater green cover.

10.96 Collectively, these reforms are streamlining approvals, reducing compliance costs, strengthening monitoring, and thereby aligning environmental governance with India's development and climate objectives. These reforms also represent a decisive shift toward trust-based environmental governance.

Box X.10: Balancing Growth and Green Initiatives...Some evidence

Tamil Nadu, Kerala, and Andhra Pradesh have not only enhanced their business environments by integrating sustainability measures but have also, through the Business Reform Action Plan (BRAP), reduced the time taken for environmental clearances.

¹¹⁹ Ministry of Environment, Forest and Climate Change, Government of India. (2025, October 29). *Rationalizing the requirement of green belt/green cover for industrial estates/parks and individual industries based on pollution potential* [Office memorandum/Notification]. Retrieved December 28, 2025, from <https://tinyurl.com/ytyd5988>.

Kerala has, under the BRAP 2024,¹²⁰ streamlined business registration, digitised land and tax processes, and simplified environmental clearances. The state also advanced renewable energy adoption, carbon-neutral gram panchayats, and waterbody rejuvenation.¹²¹

Tamil Nadu has enhanced the ease of doing business through single-window clearances, digitised approvals, and land reforms, while promoting solar parks, district-level decarbonization plans, and energy efficiency programs. The Tamil Nadu Pollution Control Board actively monitors industrial effluent treatment systems and supports the establishment of Common Effluent Treatment Plants for clusters of small-scale and pollution-intensive industries, such as tanneries and textile units, to ensure treated wastewater meets prescribed environmental standards before discharge.¹²² Andhra Pradesh, in BRAP 2024, implemented single-window industrial clearances, online land registration, and e-environmental approvals. Recently, Andhra Pradesh has expanded its Online Consent Management & Monitoring System, allowing firms to apply for consents and track approvals digitally, reducing delays and enhancing data transparency between industry and the Pollution Control Board.¹²³

Conclusion: Towards a green, resilient and competitive India

10.97 According to the synthesis report by UNFCCC, which compares the NDC targets of countries with their self-reported emission projections for 2030, none of the developed countries were on track to meet their 2030 emission reduction targets with their mitigation actions in 2020,¹²⁴ despite having an abundance of means of implementation – finance and technology. Even the new NDCs for 2035 of major developed economies show only limited enhancement in their climate mitigation goals, backpedalling on their commitments under the Paris Agreement, despite the fact that these countries reached their peak emissions decades ago, vis-à-vis developing countries like India. To add to the weakening of global efforts against climate change, the climate finance commitments to the developing countries continue to remain diluted. The clear signals from the global north, which are dithering on their own climate action, warrant that India must place adaptation centre stage in India's climate action story to ensure that development gains are not lost.

¹²⁰ Department for Promotion of Industry and Internal Trade (DPIIT): Business Reform Action Plan (BRAP) Reports, Government of India.

¹²¹ NITI Aayog. (2024). SDG India Index 2023-24. Government of India. Retrieved December 28, 2025, from <https://tinyurl.com/4ca4ku4a>.

¹²² Tamil Nadu Pollution Control Board. (2025). Policy Note 2025–2026: Environment and Climate Change [PDF]. Government of Tamil Nadu. Retrieved December 28, 2025, from <https://tinyurl.com/3beumy3>.

¹²³ Andhra Pradesh Pollution Control Board. (n.d.). Online Consent Management & Monitoring System (APPCB OCMMS). https://apocmms.nic.in/APPCB/?utm_source.

¹²⁴ Para 190 in United Nations Framework Convention on Climate Change. (2023, June 21). Compilation and synthesis of fourth biennial reports of Parties included in Annex I to the Convention: Revised report by the Secretariat, Addendum (FCCC/SBI/2020/INF.10/Add.1/Rev.2) [Advance version]. Retrieved December 28, 2025, from <https://tinyurl.com/2kur6h8j>.

10.98 The chapter underscores that climate action is no longer an environmental add-on but a core component of India's development strategy. For India, adaptation has rightly emerged as the cornerstone of climate-resilient growth, delivered primarily through public investment, state-led planning, and community institutions. The evolution of SAPCCs from mitigation-oriented documents to development-integrated adaptation frameworks reflects a growing recognition that resilience outcomes—such as water security, stable livelihoods, disaster risk reduction, and ecosystem services—are best achieved through locally grounded and institutionally embedded interventions.

10.99 India has not yet reached its peak energy demand, and ensuring energy access, affordability, and security remains central to our development pathway. India's mitigation strategy reflects pragmatism rather than prescriptiveness. Rapid growth in renewable energy, advances in energy efficiency, expansion of nuclear power, development of green hydrogen, strengthening of carbon sinks, and the operationalisation of carbon markets demonstrate how mitigation can be aligned with energy security, industrial competitiveness, and job creation. India's approach acknowledges the limits of one-size-fits-all transitions and emphasises system reliability, affordability, and domestic manufacturing capabilities.

10.100 Climate finance remains the binding constraint. The gap between global capital availability and climate investment needs in developing countries reflects structural weaknesses in the international financial architecture rather than a lack of ambition or bankable projects. India's experience highlights the importance of robust domestic financial markets, strong development banks, effective municipal finance, and credible regulatory frameworks. However, domestic resources alone cannot meet the required investment scale. Reforms to multilateral development banks, the greater use of risk-sharing and blended finance, recalibration of credit rating practices, and predictable concessional finance are essential to lowering the cost of capital and attracting private investment.

10.101 Raising climate ambition in India—especially on mitigation—without corresponding support in finance and technology is neither realistic nor equitable. A credible and orderly transition from fossil fuels depends on the timely availability of reliable, non-fossil energy sources such as nuclear power, alongside a well-defined peak-emissions pathway. There could be a push towards establishing a credible national platform, backed by predictable and adequate international finance, that can help us develop climate projects, enhance their bankability, and enable the required climate action. This, in some ways, would balance climate action ambitions with the need for secure energy in the absence of viable scalable options. In the context of such a national platform, the focus could be on scaling up innovative mitigation and adaptation

technologies, as well as on research and development, led through collaboration with institutions and partners from developed countries.

10.102 Environmental regulation is increasingly becoming a facilitator of sustainable growth. India's thoughtful transition toward risk-based regulation, market-oriented instruments, digital compliance systems, and a framework of trust-based governance highlights the potential for environmental protection to coexist harmoniously with improved business operations. Insights from various Indian states indicate that factors such as regulatory predictability, administrative capacity, and the adoption of technology are crucial

10.103 Taken together, the chapter highlights that India's climate pathway is anchored in development realism. Climate action is pursued not through abrupt transitions or externally imposed models, but through adaptive institutions, domestic capacity building, and policy coherence. As climate risks intensify and global transitions accelerate, India's experience offers a development-centric framework, one that integrates resilience, competitiveness, and sustainability, demonstrating that economic growth and environmental stewardship can advance together rather than in opposition.
